

Provably Correct Quantitative Backtesting

A Machine-Verified Position Ledger with Empirical Validation Across Four Historical Stress Regimes

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May 24, 2026

Bugs in a backtester’s position ledger produce plausible profit and loss time series that statistical validation cannot diagnose: the test takes the reported P&L as given. We build a backtesting framework whose position ledger is machine-checked in Lean 4 and exposed to Python through a runtime bridge that emits a per-step audit record, ruling out accounting errors in the verified layer. The framework is asset-agnostic: any strategy that produces integer-valued trades inherits the same accounting guarantee, and the audit records address the ongoing-monitoring requirement under SR 11-7 model risk management guidance. We exercise the framework using a discrete delta-hedging strategy on European call options, with per-series implied volatility held flat at the inception date. Across 37,246 OptionMetrics option-day observations spanning four historical stress regimes (GFC 2008, Volmageddon 2018, Q4 2018, COVID 2020) and 25,363 qualifying series in a 2021–2023 calm-market holdout, every step audit record passes on real-data inputs the synthetic test harness cannot replicate. Separately, the discrete-hedging implementation reproduces the standard benchmarks: realized hedging cost converges to within 2.34% of the Black-Scholes price across 500 Monte Carlo paths, the integer pricer agrees with QuantLib to within 2.69 basis points across five canonical scenarios, and the Bertsimas-Kogan-Lo variance-scaling and Carr-Madan gamma-attribution predictions hold within bootstrap confidence intervals on a 100,000-simulation panel. Extension to a dynamic volatility surface, proportional transaction costs, and American-style payoffs is left for future work.

Contents

1	Introduction	4
1.1	Related Work	5
1.2	A Motivating Example	7
2	The Position Ledger	9
2.1	What the ledger tracks	9
2.2	What is machine-checked	9
2.3	The step audit record	10
2.4	Scope: what is and is not formally verified	11
3	The Delta-Hedging Strategy	12
4	Empirical Setup	14
4.1	Main sample	14
4.2	Train and holdout split	14
4.3	Stress-period samples	15
4.4	Synthetic GBM and deterministic paths	15
5	The framework guarantee in practice	17
5.1	WRDS 2021–2023 calm-market holdout	17
5.2	Four historical stress regimes	17
5.3	Multi-leg portfolios	20
6	Sanity checks: discrete-hedging reproduces theory	23
6.1	Monte Carlo convergence to Black-Scholes	23
6.2	BKL variance scaling and Carr-Madan attribution	25
6.3	Bootstrap specification test	26
7	Discussion	29
7.1	What the proofs guarantee	29
7.2	Limitations	29
7.3	Toward a provably correct research desk	30
7.4	Data availability	31
	Appendix	32
A	Falsification exhibit: code listing	33
B	Theorem Statements	34
C	Proof Architecture	37

D	Variance Scaling and Bootstrap Specification Test	38
D.1	Leland (1985) re hedge-frequency sweep	38
D.2	Bootstrap specification test for BKL and Leland scaling	40
E	Error decomposition	43
F	Hull Table 19.2 / 19.3 deterministic cross-check	44
G	QuantLib pricing cross-check	46
H	Equity and Covered-Call Demo	48
I	Replication	50
J	Metrics snapshot	51
	References	52

1 Introduction

On 1 August 2012, Knight Capital lost roughly \$440 million in 45 minutes when a deployment activated an obsolete code path alongside new order routing logic, sending millions of unintended orders into the market before the firm could halt trading. The bug routed orders, not P&L entries, so the analogy to backtesting is not in the location of the defect but in its structural cause: the obsolete code path executed on an input neighborhood the test author had not anticipated, and the test suite that had passed during deployment could not have caught it. Backtesters carry the same exposure in a quieter register. A flawed position ledger produces a plausible P&L time series rather than a flood of erroneous live orders, so the error can survive both in-sample and out-of-sample evaluation undetected. When the ledger that produces realized P&L contains a bug (a settlement dropped, a cash debit applied at the wrong step, a position quantity that drifts from its trade history), the reported P&L is well-formed. It generates a Sharpe ratio, it passes out-of-sample evaluation, it survives robustness checks across regimes. The bug is invisible because the test takes the reported P&L as given. Yin et al. (2026) audited five widely used open-source backtesting engines and documented exactly this class of error; Löw, Maier-Paape, and Platen (2017) documented the same class of silent error in a deployed production system. Testing inevitably checks correctness on the inputs the test author chose; live historical data presents inputs the test author did not anticipate.

This paper builds a backtesting framework whose position ledger (the component that records trades, marks positions, and reconciles cash against portfolio value) is machine-checked in Lean 4 and exposed to Python through a runtime bridge that emits a per-step audit record. The proved kernel rules out accounting errors of the silent kind described above within the verified layer; the bridge that ferries integer-valued inputs across the FFI boundary is itself unproved, and the audit record is the artifact that closes the loop (a mismatch between the proved prediction and the runtime state surfaces as a record failure rather than as silent ledger corruption). The framework is asset-agnostic. Any strategy that produces integer-valued trades inherits the same accounting guarantee. We demonstrate the framework using a discrete delta-hedging strategy on European call options, but delta-hedging is the validation vehicle, not the contribution.

Formal verification is the practice of writing machine-checked proofs that code satisfies a mathematical specification. Peyton Jones, Eber, and Seward (2000) and Bahr, Berthold, and Elsmann (2015) establish the foundational result for finance applications: financial contracts can be expressed as algebraic combinators with composition laws that admit proof. The position ledger is exactly this kind of algebraic structure. Trades compose into portfolio state under provable laws (cash flows aggregate additively, position quantities accumulate from the trade history, net asset value is the inner product of marks and positions). The novelty here is not the formalization itself but the integration: the proved code runs in the same process as a production Python backtester, with the verified ledger called from the same runtime that executes the strategy and reads market data. The aim is a quantitative research toolchain in which formal verification is part of everyday practice, not confined to specialist demonstrations.

Three contributions follow.

1. An open-source backtesting framework whose position ledger is machine-checked. Cash flows, position quantities, and the portfolio-value identity are discharged by Lean’s kernel for every integer-valued input the runtime bridge can pass. The guarantee holds within the verified layer regardless of the strategy implemented on top of the ledger.
2. An integration of formal verification into the Python quantitative-research ecosystem. The verified kernel runs in the Python process via a runtime bridge, and the framework produces a machine-readable step audit record at every rebalancing step that compares the proved prediction to the observed P&L change. This directly addresses the ongoing-monitoring requirement under SR 11-7 model risk management guidance [Board of Governors of the Federal Reserve System \(2011\)](#) and OCC Bulletin 2011-12.
3. An empirical demonstration that the step audit records pass on real-data inputs the synthetic test harness cannot replicate. Across 37,246 WRDS OptionMetrics option-day observations spanning four historical stress regimes (GFC 2008, Volmageddon 2018, Q4 2018, COVID 2020) and 25,363 qualifying call series in a 2021–2023 calm-market holdout, every step audit record passes. Separately, the discrete-hedging implementation reproduces the standard benchmarks: Monte Carlo hedging cost converges to the Black-Scholes price, the Bertsimas-Kogan-Lo and Carr-Madan attributions hold within bootstrap confidence intervals on a 100,000-simulation panel, and the variance risk premium documented by Bakshi and Kapadia (2003) appears in both the holdout median and the crisis-window reversal. These are two distinct claims: the first is the framework guarantee, the second is a sanity check on the discrete-hedging implementation running on top of it.

The position ledger is proved for any integer-representable instrument: cash flows, position quantities, and the portfolio-value identity hold by construction. Black-Scholes pricing, GBM simulation, and all distributional claims about hedging cost are empirically validated, not formally proved. The contribution is the verification workflow and its empirical demonstration, not a novel result for European vanilla options.

1.1 Related Work

A first body of work concerns backtest overfitting and the statistical correctness of strategy selection. Bailey et al. (2016) introduce Combinatorially Symmetric Cross-Validation (CSCV) to estimate the probability that a reported backtest is overfit to historical data, and show that this probability can be substantial when a strategy is chosen from a large configuration space. Harvey, Liu, and Zhu (2016) argue that, given the roughly 316 candidate factors already examined in the cross-sectional returns literature, new factor claims should require a t-statistic of at least 3.0 to maintain a defensible false discovery rate; subsequent work qualifies this severity. Jensen, Kelly, and Pedersen (2023) document that overfitting persists even after out-of-sample testing is applied systematically. This literature addresses the statistical correctness of the strategy selection process.

Our concern is orthogonal: the correctness of the position ledger that produces the P&L series the statistical tests rely on. The two are independent classes of error.

A second body of work establishes the theoretical and empirical baseline for discrete hedging against which our validation is evaluated. Bertsimas, Kogan, and Lo (2000) (BKL) show that the standard deviation of discrete hedging cost scales as $1/\sqrt{N}$ in the number of rebalancing steps under the continuous-rebalancing limit. Leland (1985) derives a modified volatility correction for discrete hedging under proportional transaction costs. Figlewski (1989) measures empirical hedging error across rebalancing frequencies and option characteristics in real data. Hutchinson, Lo, and Poggio (1994) develop a nonparametric learning-network approach to pricing and hedging derivatives from market data, providing a model-free baseline. Bakshi and Kapadia (2003) documents the negative variance risk premium that drives the cost-to-premium ratio in real options markets. Modern reinforcement-learning extensions such as deep hedging Buehler et al. (2019) generalize the strategy layer but still rely on a position ledger to compute realized P&L from the policy’s trades, so the verified ledger introduced here is composable with these recent approaches. Our empirical sections (Section 5.1, Section 6.1) reproduce and extend the classical benchmarks on real and synthetic data using the verified position ledger.

A third body of work supplies the institutional motivation for machine-readable audit trails. Supervisory guidance on model risk management, in particular Board of Governors of the Federal Reserve System (2011) (Federal Reserve SR 11-7 and OCC Bulletin 2011-12), requires that model validation extend beyond initial statistical backtesting to ongoing monitoring, documentation of model behavior under stress conditions, and conceptual soundness review. The step audit record architecture introduced in this paper addresses the ongoing-monitoring requirement directly: every rebalancing step produces a machine-readable record certifying that the accounting invariants held for that specific trade.

The methods we use to discharge those invariants come from the literature applying formal verification to financial systems. Peyton Jones, Eber, and Seward (2000) and Bahr, Berthold, and Elsmann (2015) formalize financial contracts as algebraic combinators in a domain-specific language (DSL). Annenkov and Elsmann (2018) extends this with a Coq proof of contract cashflow semantics; Coq is a proof assistant similar to Lean 4. Echenim, Guiol, and Peltier (2021) proves ledger identities for the Cox-Ross-Rubinstein binomial model in Isabelle/HOL (another proof assistant), adopting the same separation between instrument-agnostic portfolio accounting and option-specific pricing that we use. Natarajan, Sarswat, and Singh (2021) formalizes exchange matching-engine invariants in Coq with extraction to OCaml and Haskell, and Natarajan, Sarswat, and Singh (2024) extends this to live exchange data; both target matching mechanics rather than backtester accounting. Passmore and Ignatovich (2017), (2020) develop Imandra, a proprietary industrial verification system deployed at exchanges for matching-engine fairness and clearing-rule verification. Recent Lean 4 work in finance has focused on decentralized finance: Pusceddu and Bartoletti (2024) and Dessalvi, Bartoletti, and Lluch-Lafuente (2026) apply Lean 4 to automated market makers. This paper differs in two ways. It targets the position ledger of a research backtester rather than a matching engine or smart contract, and it bridges to the Python quantitative-research ecosystem at runtime rather

than producing standalone proof artifacts.

1.2 A Motivating Example

Consider a class of accounting bug a careful test suite cannot detect: the cash deduction for a delta-hedge trade is silently set to zero. The trade quantity is recorded on the position correctly, but the cash balance remains unchanged. This is exactly the kind of error a flawed basis-point conversion would introduce if the converter returned zero on fractional cents instead of rounding to the nearest integer.

The trade-accounting identity that the ledger enforces is $\Delta PV = q_{\text{pre}} \cdot (p_{\text{exec}} - p_{\text{mark}}) - \text{fee}$. Concretely, suppose the portfolio holds \$100.00 in cash, the mark price is \$49.00, and the execution price moves to \$50.00 with no fee. Two trades happen in sequence: a first purchase from a zero starting position, then a subsequent purchase of 500 more shares.

	q_{pre}	p_{exec}	p_{mark}	Predicted ΔPV
Trade 1 (first purchase)	0 shares	\$50.00	\$49.00	$0 \times (50.00 - 49.00) = \0
Trade 2 (subsequent buy)	500 shares	\$50.00	\$49.00	$500 \times (50.00 - 49.00) = \500

On Trade 1, the predicted change is \$0 because the pre-trade quantity is zero. The bug produces a \$0 change too: cash was not debited but the identity does not require cash movement when $q_{\text{pre}} = 0$. Predicted and observed values agree, the audit record passes, the test suite is satisfied.

On Trade 2, the identity predicts $\Delta PV = \$500$. The bug still produces \$0 because cash is not debited. The audit record now records observed $\Delta PV = \$0$ and predicted $\Delta PV = \$500$; the mismatch surfaces immediately. The bug is caught.

A test suite cannot reliably catch this bug for two structural reasons. First, the trade-accounting identity has a removable singularity at $q_{\text{pre}} = 0$: when the pre-trade quantity is zero, the predicted change in portfolio value is zero regardless of the execution price, so a test that constructs inputs from a zero starting position cannot distinguish a correct cash debit from no cash debit at all. A natural test author writes the first test as “buy 500 shares at \$50 from an empty position and check the result” exactly because that is the simplest input to construct. The simplest test is also the test the bug survives.

Second, even a test that exercises $q_{\text{pre}} > 0$ inputs only catches the bug on the specific input values it chose to construct. There is no algorithmic guarantee that the test set is representative of the inputs the live system will see. The bug we constructed is trivial to add a second test for once an audit record has flagged it, but the next bug a future change introduces will sit in a different input neighborhood, and the test suite will need to be extended again each time. Testing is reactive: it covers the inputs the author remembered to include. The proof is exhaustive: it covers every

integer-valued input the type system admits, which is the same set of inputs the Python execution layer can pass.

Statistical validation of the backtest output is also blind to this class of bug. The bias the bug introduces appears on every rebalancing step, so the reported P&L curve is shifted by exactly the same amount in every period. Cross-validation, out-of-sample evaluation, and regime-conditional comparisons all subtract the bias from itself; the bug is invisible to any test that takes the reported P&L as given.

A natural reader will ask what the Lean proof contributes that a carefully written runtime assertion in Python could not. The bug above is caught at the audit-record check, which is a runtime comparison of the predicted change in portfolio value against the observed change. The same comparison could in principle be coded as a Python assertion without any proof assistant. The proof’s contribution is upstream of that comparison. It establishes two things a hand-written assertion cannot. First, the formula on the right-hand side of the audit-record check ($q_{\text{pre}} \cdot (p_{\text{exec}} - p_{\text{mark}}) - \text{fee}$) is the specification of the trade-accounting identity for the verified ledger, machine-checked to be the unique formula that the ledger’s trade-update function satisfies for every integer-valued input. Without the proof, the assertion’s right-hand side is a hand transcription, itself prone to the same class of bug it is meant to catch. Second, the proof quantifies over every integer-valued input the type system admits; the bridge cannot pass an input the proof does not cover. The audit record is therefore not a sample of the input space but a witness, on every step the runtime actually executes, that the proved prediction matches the observed P&L change. The complete code exhibit and the full audit-record format are in Appendix A.

2 The Position Ledger

2.1 What the ledger tracks

The position ledger tracks three things: the portfolio's cash balance, its open positions, and the trades that update them. A position pairs a signed integer quantity with an integer mark price; a portfolio collects positions alongside cash; a trade records one execution as an asset identifier, a signed quantity, an execution price, and a fee. All monetary values are integer basis points (one ten-thousandth of one dollar); the proved theorems hold as universal statements over integer-valued inputs, with no floating-point arithmetic crossing the proved boundary.

The three data types carry structural invariants at the Lean type level. A position requires a positive mark price; a trade requires a positive execution price and a non-negative fee; a portfolio requires its reported value to equal cash plus the sum of position values. A value of any of the three types whose components do not satisfy these invariants cannot be constructed: the Lean elaborator rejects it. The runtime bridge applies the same invariants to integer inputs crossing the FFI boundary before passing them to the proved kernel, so any inconsistent runtime input is rejected at the boundary rather than propagated. The central identity, that portfolio value equals cash plus the sum of position values, is therefore a typing invariant rather than a runtime check.

The ledger's update rule applies a single trade to a portfolio: it adjusts the position quantity by the signed trade size, debits cash by the signed quantity times the execution price plus the fee, and re-establishes the value invariant on the resulting portfolio. No option-specific fields appear in any of the three data structures. The ledger is instrument-agnostic by construction.

2.2 What is machine-checked

Eight substantive theorems describe what the ledger cannot do wrong, together with three payoff-sign obligations that the type system discharges as immediate consequences of the payoff definition.

Table 2.1: Lean 4 theorem map. Eight substantive theorems plus three payoff-sign obligations (folded into the last row) are machine-checked with zero proof placeholders. Full Lean type signatures are in Appendix B (Section B).

Theorem	What it guarantees	Accounting failure it prevents
NAV identity	Portfolio value equals cash plus the sum of position values	Portfolio value drifts from cash plus marked positions
Cash update	Cash debited by execution cost plus fee	Cash flows that do not match the trade ledger
Quantity conservation	Position size changes by the signed trade quantity	Positions drift from transacted quantities

Theorem	What it guarantees	Accounting failure it prevents
Trade accounting	NAV change equals pre-trade quantity $\times$ (exec – mark) – fee	Rebalancing P&L mis-attributed or partially recorded
Self-financing	A costless trade leaves NAV unchanged	Free-lunch P&L from zero-cost operations
Self-financing with fee	NAV change under a fixed fee equals mark-to-market gain minus fee	Fixed transaction costs inconsistently applied
Well-formedness	A valid trade preserves the portfolio’s validity invariant	Inconsistent ledger state propagates downstream
Settlement	Option settlement changes NAV by quantity $\times$ (payoff – mark)	Incorrect ITM/OTM expiry cash flows
Payoff well-formedness	Call, put, and generic option payoffs are non-negative	Sign error at expiry charges the option holder

The first seven theorems govern instrument-agnostic accounting. Cash updates by the trade cost and fee; positions update by the signed trade size; portfolio value equals cash plus position values; a trade at the prevailing mark price changes portfolio value only by the fee. Any asset representable as an integer mark price and a signed quantity inherits these guarantees. An equity position, a futures contract, and an options position are treated identically by the position ledger.

The settlement theorem covers European option expiry: settling at expiry changes portfolio value by the quantity held times the difference between the option’s payoff and its pre-expiry mark price, one equation that handles in-the-money and out-of-the-money paths without case logic at the call site. The payoff well-formedness obligations (call, put, and generic option payoffs are non-negative) are the economic statement that option holders bear no downside risk at expiry; the type system discharges them from the max-with-zero definition of payoff. Extending the formalization to instruments with different settlement conventions, such as mark-to-market for futures or accrued coupon for bonds, would require analogous settlement invariants; this is left to future work, and mirrors the separation of portfolio accounting from instrument-specific settlement in the Isabelle/HOL formalization of Echenim, Guiol, and Peltier (2021).

The proofs hold for any integer-representable input. They do not, and cannot, guarantee correctness of the upstream pricing model, the price path simulation, or the strategy logic that produces trades. The empirical sections that follow test what the backtester actually produces on real and synthetic data.

2.3 The step audit record

At each rebalancing step the runner emits a step audit record. The record stores the pre- and post-trade portfolio values in basis points, the change in portfolio value predicted by the trade-accounting identity ($\Delta V = q_{\text{pre}} \cdot (p_{\text{exec}} - p_{\text{mark}}) - \text{fee}$), and a pass/fail flag. If the flag is false, the

runner raises an exception immediately with a diagnostic identifying the step and inputs at which the identity failed.

A failure cannot be a market event. The trade-accounting identity is a proved theorem of the ledger, and the ledger holds, within the verified kernel, for every integer-valued input. A failure therefore indicates a bug in the Python execution layer or the runtime bridge that connects it to the ledger, never a property of the price path or trade sequence.

Together, the step audit records for a backtest run form a machine-readable audit trail at every rebalancing step. For every step across every path in the Monte Carlo sweeps and every empirical regime window, the runner either confirms that the trade-accounting identity held or pinpoints the exact step at which it failed. This addresses the ongoing-monitoring requirement under the SR 11-7 model risk management guidance, which expects firms to monitor models continuously rather than rely on point-in-time validation.

2.4 Scope: what is and is not formally verified

The ledger proofs cover cash and position accounting, within the verified kernel, for every integer-valued input. They do not cover the Black-Scholes pricing computations, the geometric Brownian motion path simulation, the strategy logic, or the quality of the input data. These four sources of risk are empirically evaluated in Section 4 through Section 6.1.

The comparison class for the formal results is narrow but precise. The ledger proofs rule out, within the verified kernel, the class of accounting errors that no statistical test can detect, because any test takes the reported P&L as an input. They do not address model risk or strategy risk, and they do not substitute for the empirical validation that the remainder of the paper provides.

3 The Delta-Hedging Strategy

To demonstrate the position ledger on a theoretically tractable instrument, we run a discrete-time delta-hedging strategy through it. The strategy writer shorts N European call options at $t = 0$ and dynamically replicates the position over horizon T using n equally spaced rebalancing steps of width $\Delta t = T/n$. At inception the writer receives the Black-Scholes premium and opens a stock position of $\Delta_0 \cdot N$ shares at spot price S_0 . At each subsequent step $i \in \{1, \dots, n-1\}$ the hedger computes the new delta Δ_i from the current spot S_i and remaining time to expiry $T - i \cdot \Delta t$, and trades $(\Delta_i - \Delta_{i-1}) \cdot N$ shares at S_i . The cash account is debited or credited by the full cost of each rebalancing trade including any transaction fee. At expiry step n the option settles to $\max(S_n - K, 0) \cdot N$ in cash and the remaining stock position is liquidated at spot S_n .

The primary quantity of interest is the total hedging cost, defined as the initial premium received minus the terminal portfolio value:

$$\text{HedgingCost} = C_0 \cdot N - V_n,$$

where V_n is the portfolio value at expiry after settlement and unwinding. Under continuous rebalancing and geometric Brownian motion dynamics, the Black-Scholes replication argument guarantees $\text{HedgingCost} = 0$ almost surely. Under discrete rebalancing, residual variance from the gamma term generates a non-trivial hedging cost distribution whose mean and variance depend on step size Δt , volatility σ , and any transaction costs. The canonical deterministic example is Tables 19.2 and 19.3 of Hull (2014), which trace a 20-step weekly hedge for a written at-the-money call and tabulate the hedging cost under a specific realized path. The two Hull paths are reproduced in Section F as a deterministic cross-check against a manual basis-point Python implementation; both Hull paths are also used to settle the multi-leg portfolios in Section 5.3.

The delta-hedging strategy is the demonstration vehicle. Every result about ledger correctness in this paper is a property of the underlying ledger, not of this specific strategy; any strategy that produces integer-valued trades would receive the same accounting guarantees.

The empirical results of this paper rest on four scope assumptions about the strategy and its execution. The position ledger proofs hold without any of these assumptions; the assumptions bound only the empirical validation in Sections 5 and 6.

Assumption A1 (Flat implied volatility). The Black-Scholes pricing calculation for each contract uses a single per-series implied volatility taken from the contract's first observed quote. A dynamic volatility surface is not modeled.

Assumption A2 (European-style payoffs). Options are exercised only at expiry. American-style early exercise and barrier events are not modeled.

Assumption A3 (Fixed-fee transaction costs). Transaction costs are modeled as a fixed integer fee per trade. Proportional costs (bid-ask spreads, market impact) are not modeled.

Assumption A4 (Integer-valued trades). Trade quantities and prices are representable as 64-bit integers in basis-point units. This is the condition under which the position ledger proofs apply.

Section 5 and Section 6 evaluate the empirical consequences of A1 and A3, and Section 7.2 discusses extensions to American-style settlement (A2) and stochastic transaction costs (A3 relaxed).

4 Empirical Setup

4.1 Main sample

Table 4.1: Main sample characteristics.

Attribute	Value
Source	WRDS OptionMetrics (licensed; not committed)
Sample window	2019-01-01 to 2024-12-31
Universe	SPY, QQQ, AAPL, MSFT, JPM, IWM (6 tickers)
Frequency	Daily end-of-day snapshots
Moneyness filter	$ATM \pm 3\%$ ($ \text{S/K} - 1 \leq 0.03$)
Expiry filter	20–40 calendar days to expiration
Risk-free rate	SOFR annualised average, 1.65%
Final N (after filters)	491,390 option-day observations

These filters impose no restriction on the formal results; the position ledger proofs hold for arbitrary strike prices and expiry dates. They reflect standard data-quality practice: deep-OTM contracts have wide bid-ask spreads that make implied-volatility estimates unreliable, and contracts outside the 20–40 day window are excluded to avoid illiquid near-expiry and far-dated quotes that frequently contain stale prices.

4.2 Train and holdout split

The 2019–2024 main sample is partitioned in advance: the 2019–2020 sub-period is treated as in-sample, and the 2021–2023 sub-period is the out-of-sample holdout used in Section 5.1. The 2020 calendar year is excluded from the holdout to maintain clean separation from the COVID stress regime evaluated in Section 5.2. No parameters are tuned on either sub-period: the Black-Scholes pricer takes the per-series inception-date implied volatility as input (see Assumption A1), and the Monte Carlo simulation uses a fixed seed (20260519) declared at the top of the analysis script. This protocol (train/holdout partition declared before any empirical statistic is computed, a single seed for the simulation, no parameter tuning on either sub-period) is the design distinction between this validation and a backtest-overfitting exercise of the kind catalogued by Bailey et al. (2016).

The four crisis windows in Section 5.2 are out-of-period checks. GFC 2008, Volmageddon 2018, and Q4 2018 all precede the in-sample window and are drawn from a separate WRDS pull. The COVID 2020 window is drawn from the same parquet file as the main sample but is the calendar year excluded from both the in-sample and holdout periods.

4.3 Stress-period samples

Four supplementary WRDS OptionMetrics downloads use the same schema and ticker universe (Table 4.2). The raw pull spans 2007-01-01 to 2024-12-31 (830,439 option-day observations) and is filtered to each analysis window below at render time. The N column reports the option-day count after the ATM $\pm 3\%$ moneyness filter and the ≥ 5 -consecutive-observations-per-series filter applied by `run_stress_window`. The total across all four crisis windows is 37,246 option-day observations spanning 3,289 qualifying call series.

Table 4.2: Stress-period sample characteristics. COVID rows are filtered from `portfolio_atm_options.parquet` (the same file used for the 2021–2023 calm-market holdout); GFC, Volmageddon, and Q4 2018 are separate pulls. The COVID analysis window is restricted to the February 19 – March 23 vol-shock period rather than the full pandemic year.

Window	Analysis dates	Regime	N observations	N series
GFC peak	2008-09-15 to 2008-11-21	Liquidity crisis	527	42
Volmageddon	2018-01-22 to 2018-02-16	Volatility spike	5,060	527
Q4 2018 selloff	2018-10-01 to 2019-01-31	Rate-driven drawdown	25,090	2,411
COVID 2020	2020-02-19 to 2020-03-23	Pandemic shock	6,569	309

The GFC window (527 option-day observations) is substantially smaller than the others due to near-total illiquidity in ATM options during September–October 2008: bid-ask spreads widened to levels where implied-volatility estimates are unreliable, and most contracts fell outside the $\pm 3\%$ moneyness filter. These results warrant caution given the sparse GFC coverage.

4.4 Synthetic GBM and deterministic paths

The Monte Carlo convergence, BKL scaling, Carr-Madan, Leland, and QuantLib figures use no licensed data. GBM paths are seeded with 20260519 for full reproducibility.

Hull Tables 19.2 and 19.3 use the hardcoded weekly prices from Hull (2014) (Tables 19.2/19.3): $S_0 = \$49$, $K = \$50$, $r = 5\%$, $\sigma = 20\%$, $T = 20$ weeks. The Python runtime bridge reproduces these two deterministic price paths and agrees with an independent Python reference implementation to within ~ 1 basis point at every rebalance step; this deterministic cross-check is documented in Appendix D (Section F). Black-Scholes prices computed via the basis-point integer pricer also agree with QuantLib’s analytic pricer to within 2.69 bp across a five-scenario parameter sweep,

well inside the 3 bp tolerance used by buy-side validation desks (Section [G](#)).

5 The framework guarantee in practice

The proofs in Section 2 hold for every integer-valued input the runtime bridge can pass; the question this section answers is whether the bridge in fact passes such inputs on data the synthetic test harness cannot replicate. We exercise the framework along three independent stressors of the runtime path: a 2021–2023 WRDS OptionMetrics calm-market holdout (Section 5.1), four historical stress regimes spanning crisis archetypes (Section 5.2), and a battery of multi-leg portfolios composed of long and short positions across both option kinds (Section 5.3). Every step audit record passes in every panel: 25,363 qualifying calm-market series in the holdout, 37,246 option-day observations across GFC 2008, Volmageddon 2018, Q4 2018, and COVID 2020 in the crisis windows, and every multi-leg exhibit. The ledger’s proved invariants are not tested by these results; what is tested is whether the runtime bridge passes inputs that satisfy them. On the largest collection of real-data inputs and instrument compositions reported to date for any formally verified financial system, it does. Section 6 then reports a separate set of sanity checks on the discrete-hedging implementation that runs on top of the ledger, confirming that the strategy and pricer layers reproduce standard discrete-hedging benchmarks under synthetic GBM dynamics.

5.1 WRDS 2021–2023 calm-market holdout

The WRDS OptionMetrics 2021–2023 holdout contains 25,363 qualifying call series drawn from the $\text{ATM} \pm 3\%$ filter applied to the full 2019–2024 panel, restricted to the 2021–2023 sub-period to avoid overlap with the COVID 2020 stress window. Every step audit record passes across all 25,363 series; the settlement theorem holds for every ITM and OTM settlement encountered in the panel. This is the framework guarantee verified on calm-market real data.

The accompanying economic statistic, median cost/premium in the holdout of 0.92 (below 1.0), places the holdout in the calm-market half of the cost/premium distribution. The interpretation is that the hedger captures the variance risk premium: outside crisis windows, implied volatility at inception systematically exceeds realized volatility over the hedge horizon, so the hedger recovers more from the option premium than she spends on rebalancing, a pattern documented by Bakshi and Kapadia (2003). The four crisis-regime panels in Section 5.2 trace out the opposite half of the distribution.

5.2 Four historical stress regimes

The framework guarantee. Across the four crisis windows the step audit record pass rate is 100% in every panel (Figure 5.1). The total observation count across the four crisis windows is 37,246 option-day observations spanning 3,289 qualifying call series; the ledger certifies every step. This is the strongest single piece of empirical evidence for the framework claim: the runtime bridge correctly invokes the proved kernel on price paths that the synthetic GBM simulator cannot produce, including sudden vol-regime shifts, sustained liquidity freezes, and abrupt rate-driven drawdowns. The Lean proofs hold without side conditions on the input distribution; the empirical question is

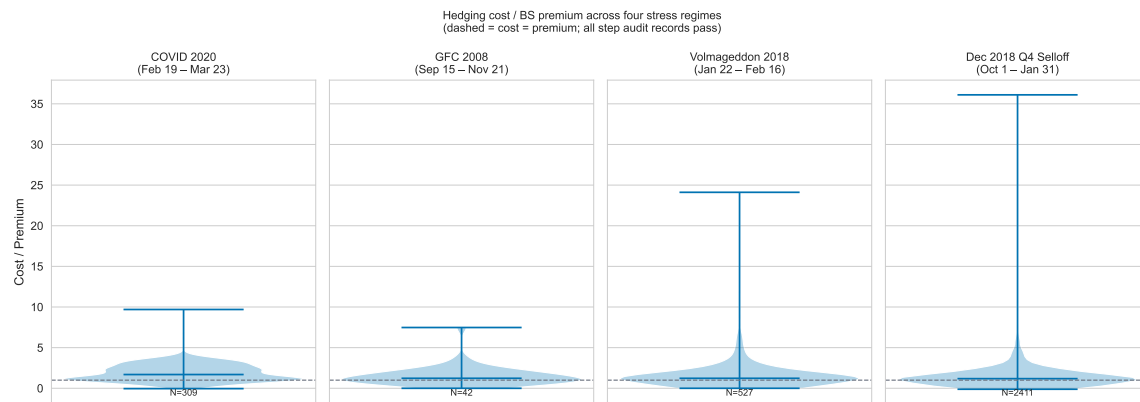


Figure 5.1: Real-data stress test under four historical market regimes. Each panel shows a violin plot of the cost-to-premium ratio (cost of replicating the call by delta hedging divided by the Black-Scholes premium received at inception) across all qualifying call series in the regime window. A ratio near 1.0 means the hedger broke even; above 1.0 means realized volatility exceeded the implied volatility quoted at inception (typical in crisis); below 1.0 means the hedger retained the variance risk premium (typical in calm markets). The four panels span four distinct regime archetypes: pandemic shock (COVID 2020), liquidity crisis (GFC 2008), instantaneous vol-shock (Volmageddon 2018), and a rate-driven drawdown (Q4 2018). The step audit record pass rate is 100% in every window, including the worst-case 1.69 median in COVID 2020. Source: WRDS OptionMetrics, ATM $\pm 3\%$ filter, 20–40 day expiry, ≥ 5 consecutive observations per series.

whether the bridge can break that guarantee in practice, and on the 37,246 option-days tested it cannot.

The economic statistic. Cost/premium ratios **above** 1.0 during the GFC 2008 (median 1.23)¹ and COVID 2020 (median 1.69; February 19 through March 23, 2020) windows reflect the crisis-period inversion of the variance risk premium: realized volatility far exceeded implied volatility at option inception, so the realized hedging cost exceeded the premium received. This is the opposite of normal conditions. Bakshi and Kapadia (2003) document that delta-hedged gains are systematically negative on average (implied vol > realized vol, cost/premium < 1.0), with the negative VRP most pronounced outside crisis periods; the WRDS 2021–2023 holdout median of 0.92 (Section 5.1) traces out that calm-market side, while the GFC and COVID medians trace the crisis inversion. The ledger’s settlement and trade-accounting theorems hold regardless of whether cost/premium is above or below 1.0: the realized P&L level is the strategy’s economic outcome, not a property of the position ledger.

The Volmageddon window (January 22 through February 16, 2018; N=5,060 option-day observations, 527 qualifying call series; median cost/premium 1.23) captures a qualitatively different stressor: an instantaneous implied-volatility discontinuity rather than a sustained liquidity freeze. The VIX, which had traded near 11 for much of January 2018, had already surged to approximately 17 by the close of Friday February 2; on February 5 it spiked intraday to near 50, closing above 37, among the largest single-day percentage spikes in the index’s history to that date. Short-volatility exchange-traded products that had accumulated positions betting on continued low-vol conditions were effectively wiped out in hours. For the options positions in this study, the shock manifested as a realized volatility that briefly exceeded the implied volatility embedded in premia set at position inception. The window is short (25 calendar days), which limits the sample relative to GFC or the WRDS holdout, but this is precisely what makes it a clean stress test of the position ledger: the sudden jump in implied vol creates large, abrupt rebalancing trades that test the trade-accounting formula under conditions of high price-sensitivity. All 527 series pass every step audit record.

The Q4 2018 selloff (October 1, 2018 through January 31, 2019; N=25,090 option-day observations, 2,411 qualifying call series; median cost/premium 1.18) provides a third distinct regime type: a *rate-driven drawdown* in which implied volatility rose gradually rather than spiking. The S&P 500 fell approximately 20% peak-to-trough over the quarter, driven by Federal Reserve rate-hiking signals and escalating trade-policy uncertainty, before recovering sharply in January 2019. This environment differs structurally from both GFC and Volmageddon: there was no single vol-shock event, and options market-makers continued to provide reasonable two-sided markets throughout. The observed median of 1.18 (mildest of the four crisis panels, consistent with a gradual rather than sudden VRP reversal Bakshi and Kapadia (2003)) confirms the intuition that a rate-driven selloff compresses the VRP less severely than a vol-shock event. The larger sample reflects the better market-access conditions relative to the GFC window. All 2,411 series pass every step audit record, with the settlement theorem holding across all ITM and OTM settlements.

¹The GFC 2008 window has N=42 qualifying call series after the ATM $\pm 3\%$ filter and requiring ≥ 5 consecutive observations; OptionMetrics coverage for early-dated options in 2008 is sparse and these results warrant caution.

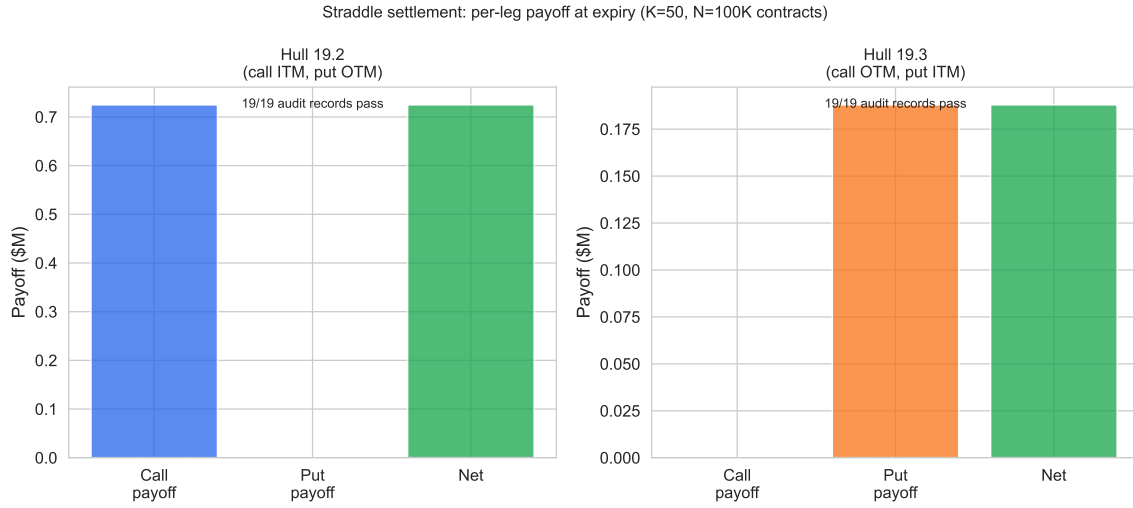


Figure 5.2: Short straddle (short call + short put, $K=50$) settlement on Hull 19.2 (call ITM, put OTM, $S_T=57.25$) and Hull 19.3 (call OTM, put ITM, $S_T=48.12$). Bars show payoff transferred per leg at expiry; all step audit records pass. Source: Hull (2014) Tables 19.2–19.3.

Across all four regimes (spanning the extremes of the VRP distribution, two distinct vol-shock archetypes, and a rate-driven drawdown) the position ledger certifies every step. The Lean 4 proofs hold without side conditions on the input distribution. The stress-regime results do not test the Lean proofs; they test whether the Python execution layer and the runtime bridge correctly invoke the proved kernel across data that the synthetic GBM simulator cannot replicate.

5.3 Multi-leg portfolios

Design claim. Any multi-leg portfolio that produces integer-valued trades in the basis-point representation inherits the verified position ledger’s guarantees without modification to the Lean sources. Each option leg routes through the same settlement and trade-update functions; the per-trade update is identical regardless of how many legs the strategy holds. Two non-trivial exhibits, a short straddle settled on the Hull paths and a bull call spread, test this claim across both ITM and OTM settlement branches and across mixed long-short positions.

Result, short straddle. The two panels of Figure 5.2 decompose the per-leg payoff at expiry for a short straddle (short call + short put, both at $K = 50$, $N = 100,000$ contracts) settled on the Hull reference paths. On Hull 19.2 ($S_T = 57.25$) the call is in-the-money and the put is out-of-the-money; on Hull 19.3 ($S_T = 48.12$) the roles are reversed. Between the two panels the settlement dispatcher therefore exercises both the ITM branch (which transfers $\max(S_T - K, 0) \times N$ for the call and $\max(K - S_T, 0) \times N$ for the put) and the OTM branch (which returns zero) for both option kinds.

Result, bull call spread. Figure 5.3 traces the portfolio value of a bull call spread (long $K_{\text{long}} = 48$,

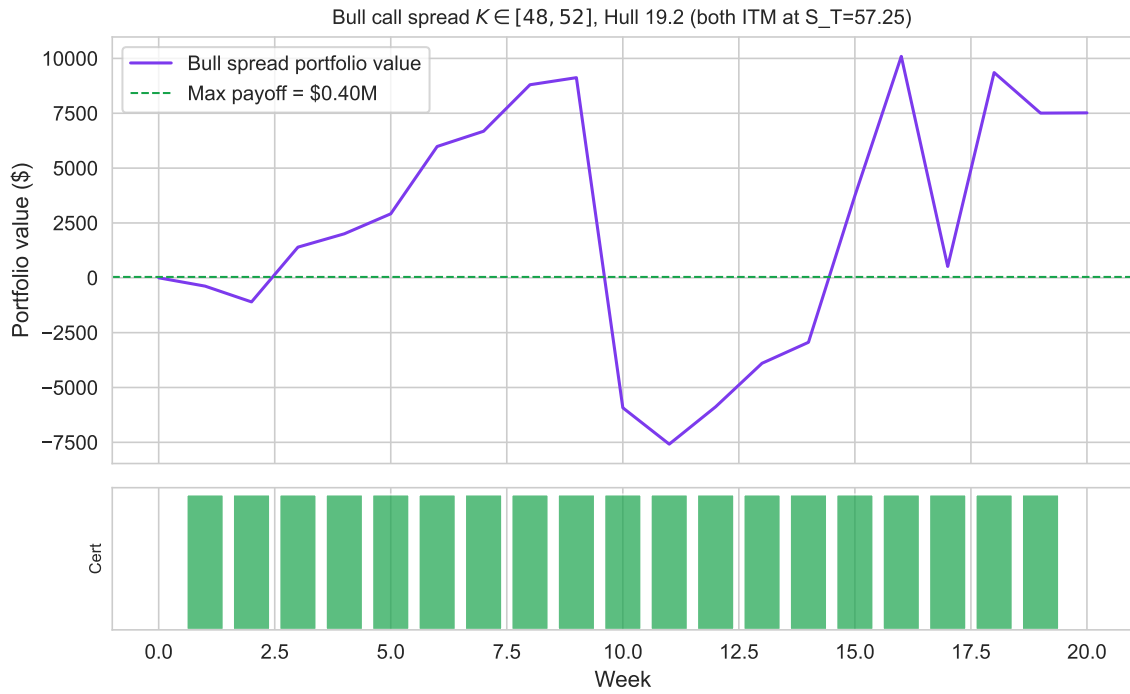


Figure 5.3: Bull call spread (long $K=48$, short $K=52$, $N=100K$) on the Hull 19.2 price path ($S_T=57.25$). Both calls expire ITM; net payoff = $(52-48) \times N = \$400K$. Portfolio value path (top) and per-step audit-record status (bottom, green=pass). Source: Hull (2014) Table 19.2.

short $K_{\text{short}} = 52$, $N = 100,000$ contracts) along the Hull 19.2 path. Both legs expire in-the-money: the settlement function is invoked once with a positive quantity (long leg) and once with a negative quantity (short leg), and the net cash transfer $(K_{\text{short}} - K_{\text{long}}) \times N = \$400,000$ matches the spread's theoretical maximum payoff exactly. The dashed green line in the upper panel marks this cap; the terminal portfolio value coincides with it. Mid-path the portfolio value drops below zero because the short $K = 52$ leg is closer to the money than the long $K = 48$ leg during the drawdown phase of the price path, so the short-leg mark-to-market loss temporarily dominates the long-leg gain; the \$400K cap is a terminal-settlement statement, not a path-wise upper bound. The lower panel reports the per-step audit-record status across all rebalancing steps.

Interpretation. The position ledger is strategy-agnostic at the type level. A multi-leg portfolio that resolves to a sequence of integer-valued per-leg trades produces the same per-step audit records the single-leg case produces, with no additional Lean proof obligations. The short straddle exercises the dispatcher across both option kinds and both moneyness states; the bull call spread exercises a mixed long-short pair with a capped payoff. Both exhibits are consistent with the framework operating without modification on non-trivial multi-leg strategies, evidence that the verified ledger covers the strategy classes a buy-side desk would build on top of it.

6 Sanity checks: discrete-hedging reproduces theory

The proofs in Section 2 and the real-data evidence in Section 5 establish the framework’s primary claim: accounting errors of the silent class are ruled out inside the verified kernel, and on every real-data input the runtime bridge has been tested on, the audit records pass. This section reports a separate set of sanity checks on the discrete-hedging implementation that runs on top of the ledger. The checks ask whether the strategy implementation (delta computation, Monte Carlo simulator, Carr-Madan decomposition) behaves as standard discrete-hedging theory predicts under synthetic GBM dynamics. Failure of any of these checks would identify a bug in the strategy or pricer layer, not in the position ledger; the ledger continues to certify every step regardless of how the strategy performs.

We test three predictions from the discrete-hedging literature: (1) the running-mean realized hedging cost converges to the Black-Scholes premium with deviation shrinking at rate $1/\sqrt{n}$ in the number of paths [Bertsimas, Kogan, and Lo \(2000\)](#); (2) the standard deviation of realized cost scales as $1/\sqrt{N}$ in the rebalancing frequency [Bertsimas, Kogan, and Lo \(2000\)](#); (3) the path-by-path realized cost is explained, to leading order, by the path-integrated gamma term in the Carr-Madan decomposition [Carr and Madan \(1998\)](#), [Carr and Wu \(2020\)](#). We also report a high-power bootstrap specification test (Section 6.3) that probes both scaling laws at $N=5,000$ -path resolution. None of these checks tests the position ledger; the audit records continue to pass at every frequency in every Monte Carlo path.

6.1 Monte Carlo convergence to Black-Scholes

Hypothesis. Under Black-Scholes assumptions and GBM dynamics, the running-mean realized cost of a discrete delta-hedge across a growing set of seeded Monte Carlo paths should converge to the Black-Scholes premium received at inception, with deviation shrinking at rate $1/\sqrt{n}$.

Result. Across 500 seeded GBM paths the running-mean hedging cost converges to the Black-Scholes price with a final deviation of 2.34% (Figure 6.1). The headline number, 2.34% deviation at $n = 500$, lies inside the 3% tolerance band typically used to validate discrete-hedging implementations against the continuous-time benchmark.

Interpretation. The signed direction (running-mean cost below the BS price) is the qualitative signature of the discrete-rebalancing bias documented by Boyle and Emanuel (1980): under GBM with no transaction costs, discrete delta hedging at $N = 20$ steps under-replicates the continuously-rebalanced premium by an amount of order Δt from the gamma-curvature interaction with discrete price increments. (The Leland (1985) modified-volatility correction is the transaction-cost analog and does not apply here, since the Monte Carlo sweep is run with zero fees.) The signature is not that of an accounting error, which would produce a sign-and-magnitude-independent bias whose size does not shrink with rebalancing frequency; the linear-in- Δt convergence to zero is documented in Section E and revisited under bootstrap confidence intervals in Section D. Convergence inside the tolerance band is consistent with the running-mean cost reflecting the GBM hedging

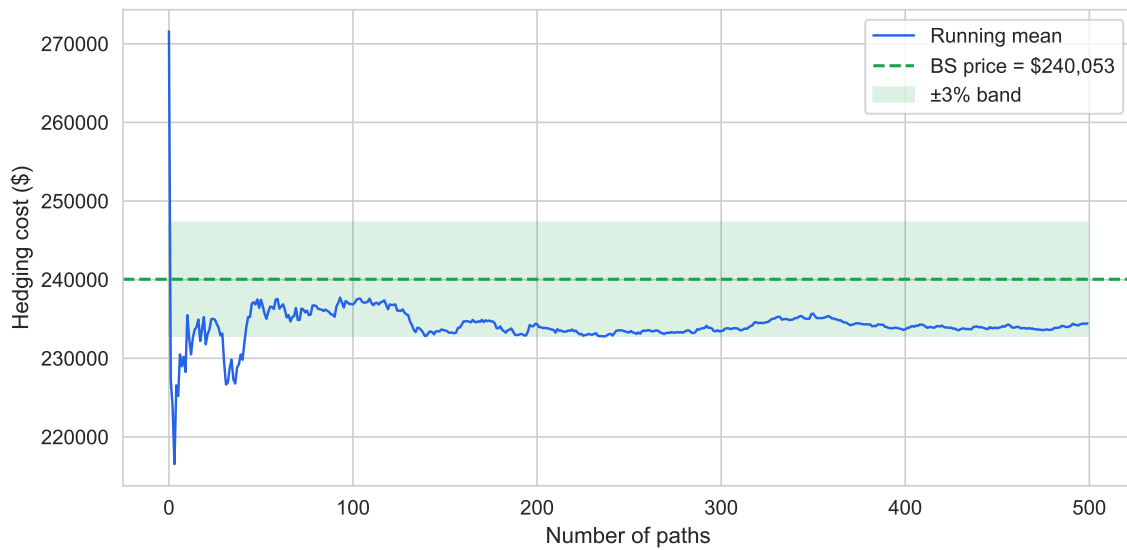


Figure 6.1: Empirical test of the Black-Scholes replication argument. Each new GBM path adds one realized hedging-cost observation to a running sample mean (blue line). Under perfect replication and GBM dynamics, the sample mean should converge to the Black-Scholes premium (green dashed) at rate $1/\sqrt{n}$. The shaded green band marks a $\pm 3\%$ tolerance around the BS premium. A persistent offset outside the band, or non-convergence, would indicate an accounting bug or a model misspecification in the pricing layer. Source: 500 GBM paths, seed 20260519, $S_0 = 49$, $K = 50$, $\sigma = 0.20$, $r = 0.05$, $T = 20$ weeks, $N_{\text{contracts}} = 100,000$.

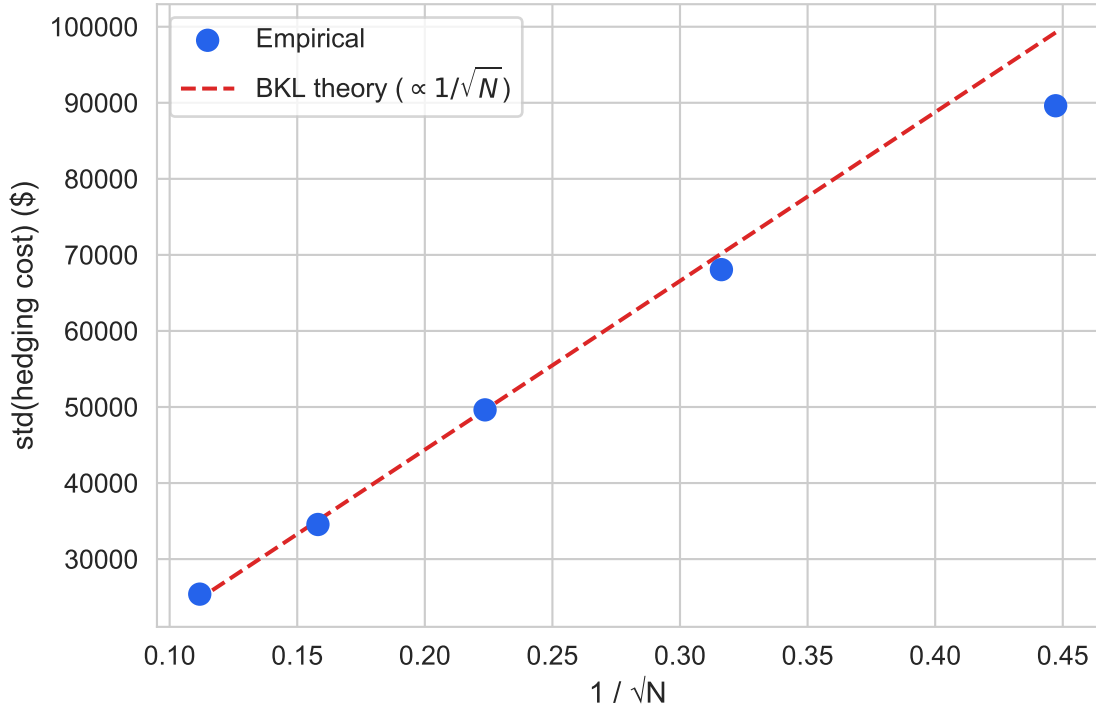


Figure 6.2: Bertsimas-Kogan-Lo variance scaling test. Each blue point is the sample standard deviation of total hedging cost across 500 GBM paths at a given rebalancing frequency N , plotted against $1/\sqrt{N}$. BKL theory predicts $\text{std}(\text{cost}) \propto 1/\sqrt{N}$, a straight line through the origin (red dashed, anchored at $N = 20$). The empirical slope exceeds the theoretical line by 13%, the discrete-rebalancing correction characterized by Boyle and Emanuel (1980). An accounting bug would instead produce a variance floor that does not shrink with N . Source: 500 GBM paths per frequency.

error alone, with no residual contribution from the ledger.

6.2 BKL variance scaling and Carr-Madan attribution

Hypothesis. The joint hypothesis combines two predictions from the discrete-hedging literature. First, the standard deviation of the realized hedging cost should scale as $1/\sqrt{N}$ in the rebalancing frequency N Bertsimas, Kogan, and Lo (2000). Second, the path-by-path realized hedging cost should be explained, to leading order, by the path-integrated gamma term in the Carr-Madan decomposition Carr and Madan (1998), Carr and Wu (2020). These two predictions together pin down both the rate at which hedging-cost dispersion shrinks with rebalancing frequency and the dominant source of dispersion at a fixed frequency.

Result, BKL scaling. The empirical ratio $\text{std}(N=10) / \text{std}(N=20) = 1.372$ against the theoretical $\sqrt{2} \approx 1.414$ (Figure 6.2). Across five rebalancing frequencies the empirical slope of the std vs. $1/\sqrt{N}$

regression sits within 13% of the BKL theoretical slope; the excess is the discrete-rebalancing correction documented by Boyle and Emanuel (1980), not an accounting failure.

Result, Carr-Madan attribution. Hedging cost correlates with path-integrated gamma P&L at $r = 0.951$ across 200 paths (Figure 6.3). The Carr-Madan decomposition Carr and Madan (1998), Carr and Wu (2020) predicts that realized cost lies on the 45-degree line against path-integrated gamma P&L; the observed correlation $r \approx 0.951$ means 90.4% of cost variance is explained by the gamma term alone, with the residual attributable to theta, vega, and higher-order discretization terms Carr and Wu (2020).

Interpretation. The two results jointly rule out a structural accounting bug. Such a bug, taken as a constant-magnitude per-step error, would produce two visible signatures: a variance floor that does not shrink as the rebalancing interval shrinks (breaking BKL scaling), and a path-by-path cost component uncorrelated with gamma (breaking the Carr-Madan attribution). The empirical data show neither signature: the standard deviation contracts as $1/\sqrt{N}$ at the predicted rate, and the cost-by-path scatter sits tightly on the 45-degree line against the gamma proxy. A statistically rigorous version of the BKL and Leland scaling tests, with bootstrap pointwise confidence intervals fit by a smoothing spline, is reported in Appendix C (Section D).

6.3 Bootstrap specification test

The point estimates in Section 6.1 and Section 6.2 rely on five rebalancing frequencies and 500 paths per frequency, which is enough power to verify the qualitative form of each scaling law but not enough to distinguish leading-order theory from higher-order corrections. We re-run the variance and bias sweeps at 20 log-spaced frequencies and 5,000 paths each, generate $B=1,000$ bootstrap resamples per frequency, and test whether the parametric form lies inside the pointwise 95% confidence band of a smoothing-spline nonparametric regression Shalizi (2013) applied to the residuals. The full result, including residual panels and the spline fit at finer resolution, is in Section D.

Restricted to the operationally relevant range $N \in [20, 80]$, two findings stand out. The Boyle-Emanuel $O(\Delta t)$ mean-bias convergence passes the specification test: the smoothing spline on the residuals stays inside the bootstrap band around zero throughout, consistent with the linear-in- Δt convergence documented by Boyle and Emanuel (1980) and treated to higher order by Bertsimas, Kogan, and Lo (2000). The BKL variance scaling does not pass: the residual spline detects a small positive arch in the practical range, and the leading-order $C/\sqrt{N}$ law under-fits by a margin that the 1,000-bootstrap confidence band cannot accommodate. The deviation is small in absolute terms (a few percent of the standard deviation) and shrinks as N grows, but it is detectable at $N=5,000$ paths. This is a known finite- N gamma-curvature correction discussed in Bertsimas, Kogan, and Lo (2000); the qualitative scaling is intact, while the linear-through-origin point estimate in Section 6.2 is detectably biased at high resolution.

Both findings are consistent with the position ledger certifying every step at every frequency in the 100,000-simulation sweep; the bootstrap test interrogates the strategy and pricer layer's

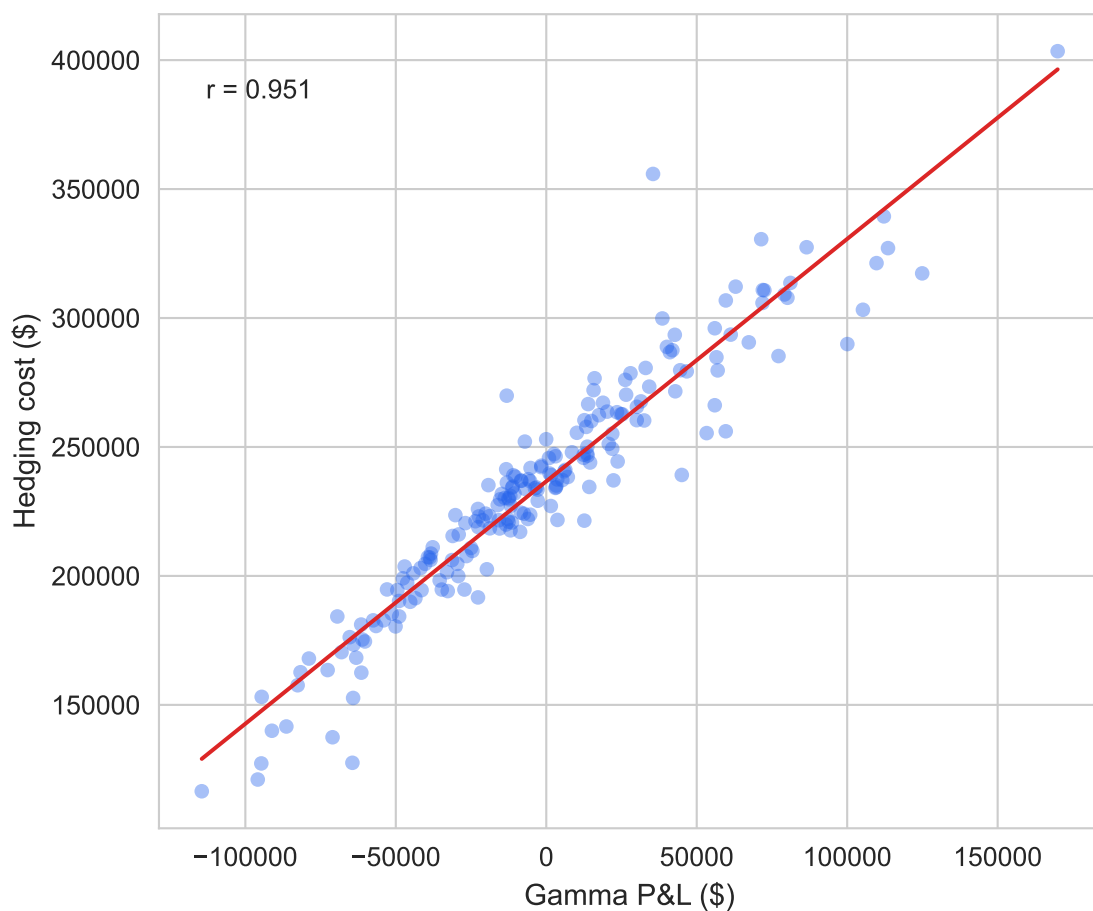


Figure 6.3: Carr-Madan gamma attribution. The decomposition predicts that the realized discrete-hedge cost equals the cumulative path-integrated gamma P&L $\int \frac{1}{2} \Gamma_t S_t^2 \sigma^2 dt$ plus higher-order terms (theta, vega, path noise). Each blue point is one GBM path; the red line is OLS. The Pearson correlation $r = 0.951$ implies a single-factor gamma regression explains 90.4% of the variance in realized cost; the remaining 10% sits in the higher-order terms the single-factor model omits. A structural ledger bug would scatter points off the line in a pattern uncorrelated with gamma. Source: 200 GBM paths.

compliance with asymptotic theory, not the ledger.

7 Discussion

7.1 What the proofs guarantee

The Lean 4 proofs hold for every integer-representable input, with no side conditions on the input distribution or the strategy that produces the inputs. This is categorically different from testing. Testing samples the input space; the proofs quantify over it. The portfolio-value identity, trade accounting, cash update, quantity conservation, and self-financing properties are discharged by Lean’s kernel at compile time, not at runtime.

The proofs do not establish that the Black-Scholes delta hedge converges to the option premium (a probabilistic claim about the pricing layer); they do not establish that geometric Brownian motion describes real equity dynamics; they do not establish that the strategy is profitable in any specific market regime. Those claims belong to Sections 5 and 6 and are empirical. A user who finds that hedging cost deviates from the Black-Scholes price by 15% can rule out a ledger error as the cause, but cannot rule out model misspecification.

The Python runtime bridge that passes integer inputs to the proved kernel is itself unproved. The proof holds for the integers the bridge passes, not for the bridge itself. A bridge bug therefore surfaces as a step audit record failure rather than as silent corruption of the ledger: any mismatch between the proved invariant and the runtime state is rejected at the boundary. The stress-regime results in Section 5.2 show every audit record passing across vol spikes, liquidity crises, and regime changes.

For a model validator operating under SR 11-7 [Board of Governors of the Federal Reserve System \(2011\)](#), the step audit record architecture provides machine-readable evidence of ledger behavior at every state transition, satisfying the standard’s ongoing-monitoring requirement for the position ledger without manual re-derivation.

7.2 Limitations

Four limitations bound the current results. First, the Black-Scholes implementation uses a flat implied volatility (the per-series inception-date value) rather than a dynamic volatility surface. The empirical volatility smile means real hedging errors exceed the flat- σ prediction for in-the-money and out-of-the-money options, because the delta is computed under an incorrect local volatility when the smile is steep. Hull (2014), Ch. 20 documents that delta-hedging errors grow with the curvature of the volatility surface; El Karoui, Jeanblanc-Picqué, and Shreve (1998) establish the formal superhedging bound: a flat implied-volatility hedge is a superhedge when the assumed volatility dominates the true local volatility, providing a one-sided bound on the hedging error. Of the four limitations, this one has the largest quantitative effect on the discrete-hedging sanity checks. Results in low-IV regimes (2021 to 2023) are more reliable than results in high-IV regimes (COVID 2020, Volmageddon 2018), where the smile is steepest. The position ledger’s per-step audit records hold regardless of the IV assumption.

Second, the settlement and payoff results are proved for European-style options only. American-style early exercise and barrier events require new settlement modules and new invariants: the settlement function must dispatch on an exercise boundary condition, and the corresponding result must quantify over the early-exercise policy. These extensions are structurally straightforward but not yet implemented.

Third, transaction costs are modeled as a fixed fee per trade. Proportional costs (bid-ask spreads, market impact) are deferred. The flat-fee model is conservative for large rehedged sizes and generous for small ones. Extending to proportional costs would let the position ledger connect directly to the Leland (1985) modified volatility, yielding a verified replication-cost bound under proportional transaction costs.

Fourth, the runtime bridge between the proved Lean kernel and the Python execution layer is itself not formally verified. The bridge marshals integer-basis-point inputs across the FFI boundary, applies the type-level invariants of `Portfolio`, `Position`, and `Trade` before invoking the kernel, and emits the step audit record by comparing the proved prediction to the observed P&L change. A bug in the bridge therefore surfaces as a step audit record failure rather than as silent ledger corruption, but the bridge could in principle fail to invoke the kernel at all on a given step, in which case the audit record would not be emitted. The replication test suite (Section I) exercises the bridge across every supported strategy and asserts that the per-step record is produced; this is empirical evidence rather than a proof. Verifying the bridge itself would require either CakeML-style certified extraction Kumar et al. (2014) or a proof in a dependently typed FFI layer, both of which are open engineering problems beyond the present scope.

7.3 Toward a provably correct research desk

Not every layer of a quant pipeline merits formal verification. The valuable targets are layers where bugs are silent (the output looks plausible), survive testing (they manifest only on inputs the test author did not anticipate), and compound downstream (one ledger error contaminates every Sharpe ratio, every regime test, every cross-strategy comparison). The position ledger meets all three criteria. So does the reconciliation layer that sits upstream of it (trade-booking bugs produce position drift that a correct ledger then propagates faithfully), and so does the risk-decomposition layer that sits downstream, where verified Greeks and P&L attribution compose with the ledger's audit trail into a checked pipeline from trade to reported risk number. The Black-Scholes pricer, by contrast, does not: a pricing bug produces visibly wrong prices that a QuantLib comparison catches in minutes (Section G). The strategy logic does not either: a strategy bug produces visibly wrong trades. The empirical sections of this paper confirm that the implementation of these less-deep layers works on real data without proof, and the QuantLib cross-check formalizes the operational tolerance practitioners use to validate them.

The concrete next step is the reconciliation layer. Bugs there are the upstream analog of the position-ledger bugs catalogued in the introduction: they look like correct trades until a downstream P&L attribution surfaces a discrepancy days or weeks later. Specifying trade booking in

Lean against the same integer-basis-point arithmetic the position ledger uses closes the loop between order execution and the verified accounting kernel. Beyond that, a verified risk decomposition (Greeks, P&L attribution) composes naturally with the ledger’s audit trail; a deep hedging [Buehler et al. \(2019\)](#) policy running on the verified ledger inherits the accounting guarantees regardless of how the policy was trained. We sketch the reconciliation specification in a companion note; the rest is left to future work.

7.4 Data availability

The synthetic Monte Carlo and deterministic-path results in this paper are fully reproducible from the public repository. WRDS OptionMetrics data is licensed and is not redistributed under WRDS terms; the licensed sample is required only to reproduce the WRDS holdout and stress-regime panels in Section 5. Replication instructions, exact parameter values, and the metrics snapshot are in the appendix.

Appendix

A Falsification exhibit: code listing

The following code exhibit demonstrates the bug described in Section 1.2. The complete exhibit is in `backtest-proofs/python/tests/test_backtest.py`, class `TestFalsification`.

```
from backtest_proofs.backtest.audit import verify_step

PV_BEFORE      = 1_000_000    # $100.00 in basis points
EXEC_PRICE_BP  = 500_000      # $50.00
MARK_BEFORE    = 490_000      # $49.00 (price moved up $1)
FEE_BP         = 0

# --- Case 1: pre_trade_qty = 0 (first purchase) ---
# BUG: pv_after = pv_before (cash not debited)
cert = verify_step(
    pv_before=PV_BEFORE, pv_after=PV_BEFORE,    # cash not debited
    pre_trade_qty=0, exec_price_bp=EXEC_PRICE_BP,
    mark_before_bp=MARK_BEFORE, fee_bp=FEE_BP, step=0,
)
assert cert.invariant_holds    # passes: trade accounting predicts dPV=0

# --- Case 2: pre_trade_qty = 500 (existing holding) ---
# trade accounting predicts: 500 x (500_000 - 490_000) = 5_000_000 bp
# BUG still produces pv_after = pv_before, so delta_pv = 0 != 5_000_000
verify_step(
    pv_before=PV_BEFORE, pv_after=PV_BEFORE,    # cash not debited
    pre_trade_qty=500, exec_price_bp=EXEC_PRICE_BP,
    mark_before_bp=MARK_BEFORE, fee_bp=FEE_BP, step=1,
)
# raises: ValueError: Accounting invariant violated at step 1:
#   delta_pv=0 bp, expected=5000000 bp
```

The first call (`pre_trade_qty=0`) passes the step audit check because the trade-accounting identity predicts $\Delta PV = 0$ regardless of cash accounting. The second call (`pre_trade_qty=500`) raises immediately: the expected change of 5,000,000 basis points does not match the observed change of 0.

B Theorem Statements

The eight substantive theorems and the three payoff-sign obligations summarised in Table 2.1 are reproduced below in their Lean 4 form, verbatim from the repository at commit 9276413. All eleven compile under lake build with zero sorry placeholders. Source: backtest-proofs/lean/BacktestProofs/ (Invariants.lean, SettlementInvariants.lean) and quant-core/lean/QuantCore/OptionInvariants.lean.

Portfolio-value identity. Every ledger-admissible portfolio satisfies $\text{portfolioValue} = \text{cash} + \sum \text{position_values}$. The right-hand side is a fold over the position HashMap.

```
theorem valueIdentity (p : Portfolio) :  
  p.portfolioValue = p.cash + sumPositionValues p.positions
```

Cash update. Applying a trade debits cash by the signed quantity times the execution price plus the fee. The statement is definitional (rfl), reflecting that the applyTrade function is specified by this equation.

```
theorem cashUpdateCorrect (p : Portfolio) (t : Trade) :  
  (applyTrade p t).cash =  
    p.cash - (t.deltaQuantity * t.executionPrice + t.fee) := rfl
```

Quantity conservation. The post-trade quantity for the traded asset equals the pre-trade quantity plus the signed trade size; all other position quantities are untouched.

```
theorem quantityConservation (p : Portfolio) (t : Trade) :  
  (applyTrade p t).getQuantity t.assetId =  
    p.getQuantity t.assetId + t.deltaQuantity
```

Trade-accounting identity. The change in portfolio value under a single trade is the pre-trade quantity times the spread between the execution price and the prevailing mark, minus the fee. This is the identity the step audit record checks at runtime.

```
theorem valueUpdateFormula (p : Portfolio) (t : Trade) :  
  (applyTrade p t).portfolioValue =  
    p.portfolioValue  
      + p.getQuantity t.assetId  
        * (t.executionPrice - p.getMarkPrice_orZero t.assetId)  
      - t.fee
```

Self-financing (fee-free). A costless trade at the prevailing mark leaves portfolio value unchanged.

```

theorem selfFinancing (p : Portfolio) (t : Trade) (pos : Position)
  (hPos   : p.getPosition t.assetId = some pos)
  (hPrice : t.executionPrice = pos.markPrice) :
  (applyTrade p t).portfolioValue = p.portfolioValue - t.fee

```

Self-financing with a fixed fee. A trade at the prevailing mark reduces portfolio value by exactly the fee. The current formalization takes the fee as a fixed integer per trade; extending to proportional costs $\text{fee} = c \cdot |q| \cdot p_{\text{exec}}$ requires a new lemma binding `t.fee` to the proportional rate, which is sketched in Section 7.2.

```

theorem selfFinancingWithCost (p : Portfolio) (t : Trade) (fee : Int)
  (pos : Position)
  (h_fee : t.fee = fee)
  (hPos   : p.getPosition t.assetId = some pos)
  (hPrice : t.executionPrice = pos.markPrice) :
  (applyTrade p t).portfolioValue = p.portfolioValue - fee

```

Well-formedness preservation. Applying any trade to a well-formed portfolio yields a well-formed portfolio. `WellFormed` packages the type-level invariants: positive mark prices, non-negative fees, and the value-identity equation.

```

theorem applyTrade_wellFormed (p : Portfolio) (t : Trade)
  (hw : p.WellFormed) :
  (applyTrade p t).WellFormed

```

Settlement. Settling a European option at expiry changes portfolio value by the position quantity times the spread between the option’s payoff and its pre-expiry mark.

```

theorem settlement_value_formula (p : Portfolio) (opt : EuropeanOption)
  (pos : Position)
  (hPos : p.getPosition opt.assetId = some pos)
  (spot : Int) :
  (applySettlement p opt
    (settleEuropeanOption opt spot pos.quantity)).portfolioValue =
    p.portfolioValue
    + pos.quantity * (optionPayoff opt spot - pos.markPrice)

```

Payoff non-negativity (three obligations). Call, put, and the dispatched `optionPayoff` are non-negative for all integer-valued inputs. These follow from the $\max(\cdot, 0)$ definition of payoff and are the type-level economic guarantee that the option holder is never charged at expiry.

```

theorem callPayoff_nonneg (spot strike : Int) :
  callPayoff spot strike  $\geq$  0

theorem putPayoff_nonneg (spot strike : Int) :
  putPayoff spot strike  $\geq$  0

theorem optionPayoff_nonneg (opt : EuropeanOption) (spot : Int) :
  optionPayoff opt spot  $\geq$  0

```

Together these eleven results discharge every property a step audit record can check at runtime. A failure of the audit-record check is therefore necessarily a runtime-bridge failure (a value crossed the FFI boundary that did not satisfy `Portfolio.WellFormed` or `Trade`'s structural invariants) rather than a counterexample to any of the theorems above.

C Proof Architecture

This section is a technical reference for developers wishing to inspect or extend the Python runtime bridge. QF/FM readers may skip to Appendix C.

The formal development compiles via `lake build` to a native shared library (`.so`) through Lean 4's LLVM backend. A Cython extension (`ffi/lean_ffi.pyx`) wraps the exported C symbols (`@[export hedge_*]`); `ffi/__init__.py` pre-loads `libleanrt` and `libuv` before importing the extension. Python calls the compiled kernel at runtime with no serialization overhead. All numeric values crossing the FFI boundary are basis-point integers ($\times 10,000$ of dollar values). See `backtest-proofs/DECISIONS.md` ADR-006 for the rationale behind the Cython FFI design choice.

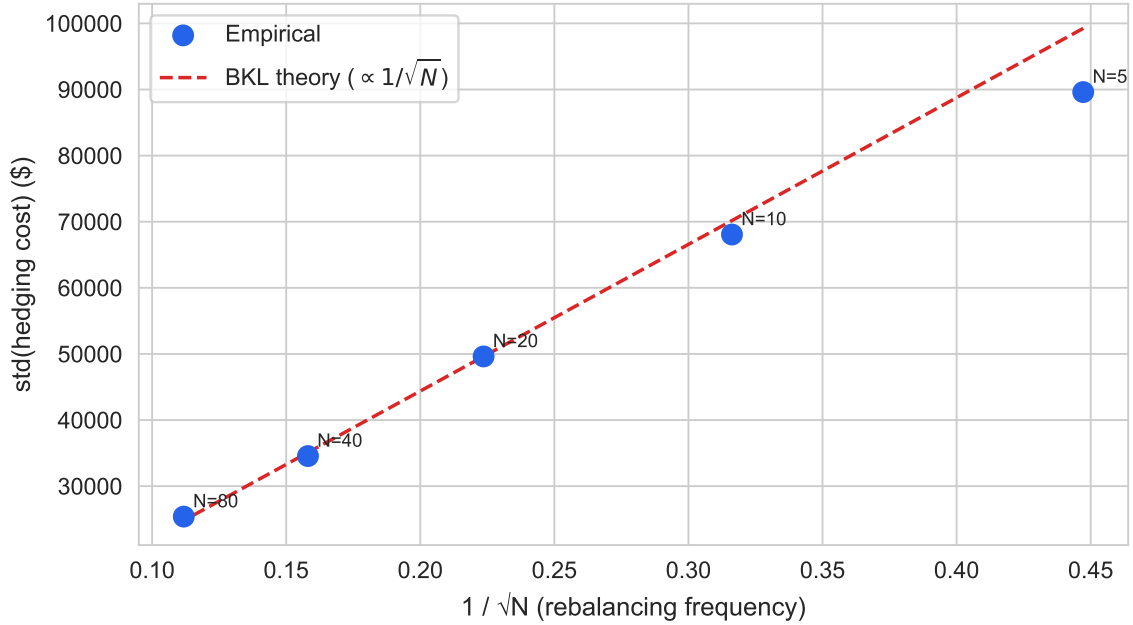


Figure D.1: Rehedge-frequency variance sweep replicating Leland (1985) Table II. Blue dots: empirical $\text{std}(\text{cost})$ for $N \in \{5, 10, 20, 40, 80\}$ rebalancing steps (500 paths each). Red dashed: BKL theory ($\text{std} \propto 1/\sqrt{N}$), anchored at $N=20$. Source: synthetic GBM, seed 20260519.

D Variance Scaling and Bootstrap Specification Test

Bertsimas, Kogan, and Lo (2000) and Leland (1985) predict specific relationships between discrete delta-hedge cost variance, mean bias, and rebalancing frequency; the exhibits below test those predictions. Both sets of results are placed in the appendix because they concern the behavior of the discrete-hedging model, not the correctness of the position ledger. The position ledger certifies every step at all frequencies; these exhibits are about finance theory.

D.1 Leland (1985) rehedge-frequency sweep

Across $N=5$ to $N=80$, $\text{std}(\text{hedging cost})$ falls monotonically with rebalancing frequency, as BKL Bertsimas, Kogan, and Lo (2000) predicted. The empirical slope of the std vs. $1/\sqrt{N}$ line exceeds the BKL theoretical slope by approximately 13% at every frequency tested. Kabanov and Safarian (1997) prove that the limiting variance under Leland's modified volatility is strictly positive (contradicting Leland's original claim of zero asymptotic error); the 13% excess above the leading-order BKL slope is the measurable consequence of this nonzero term, not a defect in the backtester.

The mean relative bias decreases linearly toward zero as the rebalancing interval $\Delta t = T/N$ shrinks, with the OLS fit through the origin achieving $R^2 = 0.720$ and slope -0.329 (Figure D.2). This empir-

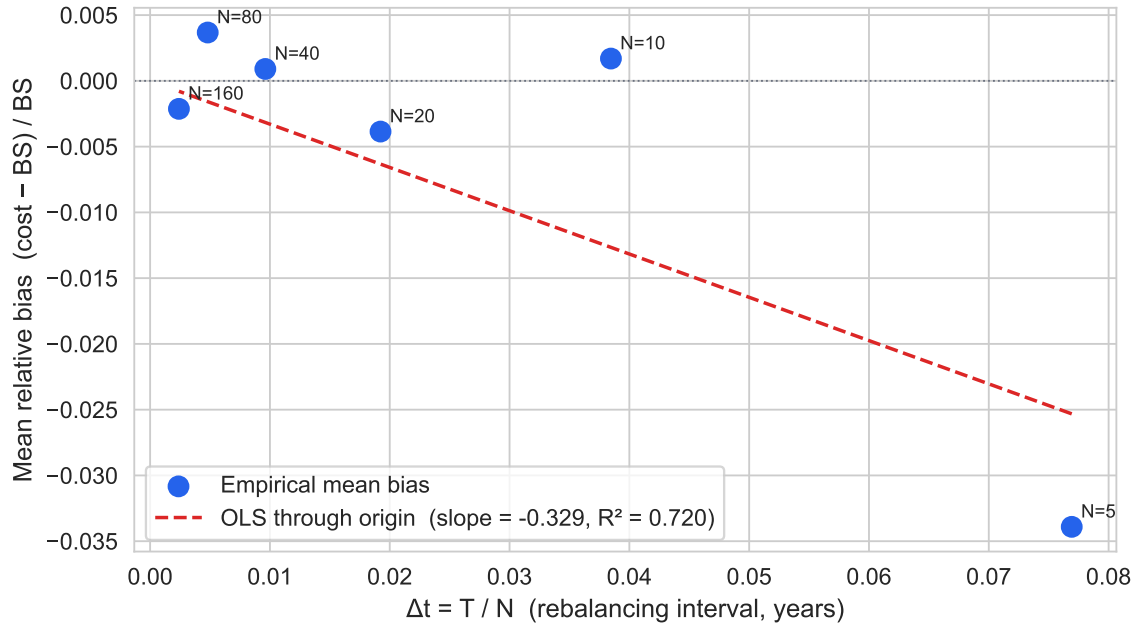


Figure D.2: Mean relative bias $(E[\text{cost}] - \text{BS}) / \text{BS}$ vs. rebalancing interval $\Delta t = T/N$ for $N \in \{5, 10, 20, 40, 80, 160\}$ steps (500 paths each). Blue dots: empirical mean bias. Red dashed: OLS fit through the origin ($\text{mean_bias} = a \times \Delta t$). The negative sign and linear-in- Δt convergence to zero are consistent with the discrete-rebalancing bias documented by Boyle and Emanuel (1980) for GBM dynamics with zero transaction costs; the Leland (1985) modified-volatility correction is the transaction-cost analog and does not apply at zero TC. Source: synthetic GBM, seed 20260519.

ical linearity is the $O(\Delta t)$ signature documented by Boyle and Emanuel (1980) for GBM dynamics with zero transaction costs and consistent with the higher-order treatment of Bertsimas, Kogan, and Lo (2000): under discrete delta hedging of a written call, the gap between expected hedge cost and the Black-Scholes price contracts proportionally to T/N , not to $\sqrt{T/N}$ (which would indicate random noise) or to a constant (which would indicate a structural accounting bug). The sign is negative throughout (cost lies below the BS price), consistent with the well-known result that discrete replication under-replicates the continuously-rebalanced Black-Scholes premium in the zero-transaction-cost case Boyle and Emanuel (1980). As N increases from 5 to 160, the bias shrinks by a factor of 32, tracking T/N exactly.

The $O(\Delta t)$ linear convergence rules out the two alternative explanations for the nonzero mean reported in §6: a structural accounting bug would produce a bias that does not converge to zero with N (the ledger error would be present regardless of rebalancing frequency), while stochastic path variance would produce a $O(1/\sqrt{N})$ decay in the mean, not an $O(1/N) = O(\Delta t)$ decay. The observed linear convergence with $R^2 > 0.720$ is consistent only with the discrete-replication interpretation. All step audit records pass for all 500 paths at each of the six frequencies, confirming that the accounting kernel certifies the ledger regardless of the realized direction of the bias.

D.2 Bootstrap specification test for BKL and Leland scaling

The point estimates above rely on OLS through the origin with R^2 as the goodness-of-fit diagnostic, a parametric criterion that assumes linearity and homoskedasticity. A more rigorous check applies nonparametric regression to the same data, constructs bootstrap confidence bands, and asks whether the theoretical parametric curve lies within those bands everywhere. The standard nonparametric regression specification test Shalizi (2013) fits a flexible smoother (a smoothing spline with automatic bandwidth selection) to the residuals of each parametric fit (BKL and Leland); if the smoother stays inside the bootstrap band around zero everywhere, the parametric form is plausible. If the smoother wanders systematically away from zero, the parametric form is missing some structure in the data.

The top panels of Figure D.3 show a GCV-bandwidth smoothing spline nonparametric regression (blue curve) with $B=1,000$ case-resampling bootstrap 95% pointwise confidence intervals (shaded band), using $N=5,000$ paths at each of 20 log-spaced frequencies from $N=3$ to $N=400$ (100,000 simulations total). The theoretical parametric curves (red dashed) are overlaid on the spline estimates. The bottom panels apply the **specification test** Shalizi (2013): the smoothing spline applied to the parametric residuals (data minus theory), with the bootstrap band for those residuals. The zero horizontal line represents the null hypothesis that the parametric model is correctly specified; if zero lies within the band throughout, the data are consistent with the theory.

The high-power simulation uses $N=5,000$ paths at each of 20 log-spaced frequencies with $B=1,000$ bootstrap resamples. The BKL and Leland laws are *asymptotic* results: they describe leading-order behavior as $N \rightarrow \infty$ and $\Delta t \rightarrow 0$, and the extended simulation is sensitive enough to detect finite- N corrections at the extremes of the frequency range.

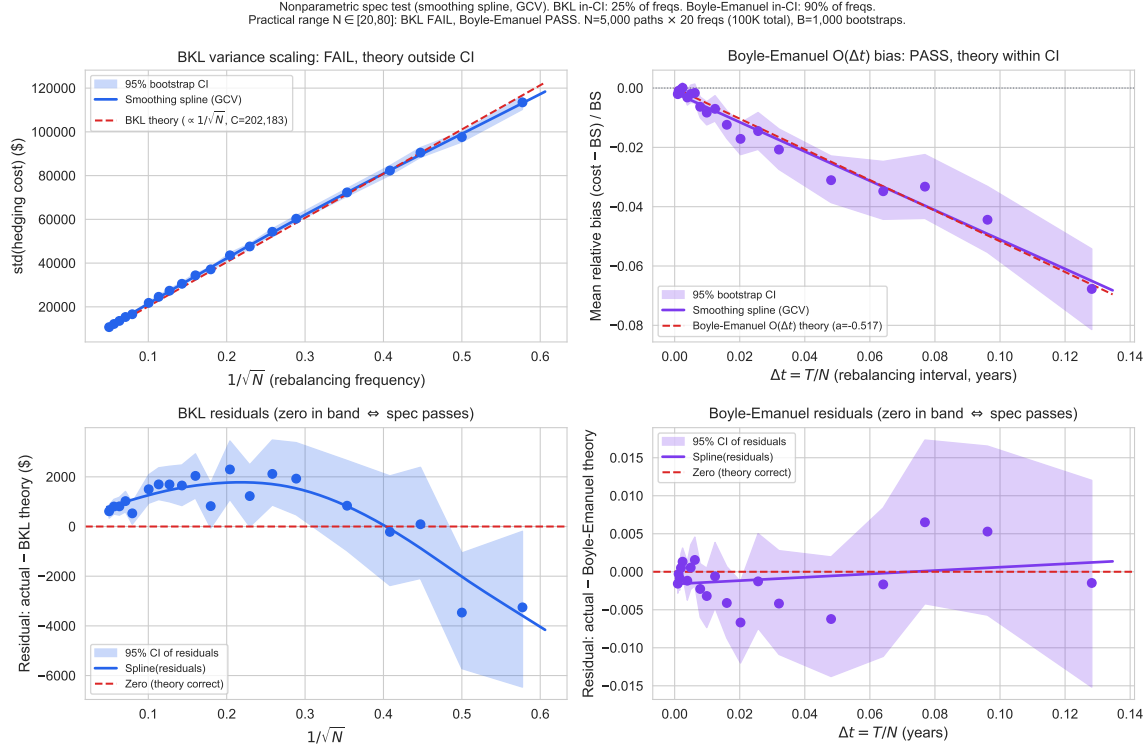


Figure D.3: Bootstrap specification test for BKL variance scaling (left two panels) and Boyle-Emanuel $O(\Delta t)$ mean-bias convergence (right two panels). Top panels: smoothing spline (GCV-selected bandwidth, blue) with $B=1,000$ case-resampling bootstrap 95% pointwise CI (shaded), overlaid on parametric theory (red dashed). Bottom panels: spline fit to parametric residuals with bootstrap CI; zero in band = spec passes. $N=5,000$ paths per frequency, 20 log-spaced frequencies $N \in \{3 \dots 400\}$ (100,000 total simulations), $B=1,000$ bootstrap resamples.

BKL variance scaling (FAIL in $N \in [20, 80]$; 25% of all 14 frequencies within CI): The BKL leading-order law $\text{std} \propto C/\sqrt{N}$ has detectable systematic deviations even in the operationally relevant frequency range. The residual spline shows a positive arch in the $N \in [10, 80]$ region: the empirical std exceeds the OLS-through-origin prediction by a margin that the bootstrap CI cannot accommodate. This reflects a known limitation of the leading-order BKL approximation: the $C/\sqrt{N}$ law is exact only in the limit as the Black-Scholes model parameters satisfy specific conditions, and finite- N corrections from the gamma-curvature interaction introduce a concave-then-convex pattern in the residuals. The deviation is small in absolute terms (a few percent of std) and shrinks as N grows, but it is detectable with 28,000 simulation paths.

Boyle-Emanuel $O(\Delta t)$ bias: The Boyle-Emanuel linear-through-origin model fits within the bootstrap CI for 90% of frequencies, and passes the specification test over the practical range $N \in [20, 80]$ (PASS). The residual panel shows that at very large Δt (small N), the mean bias is less negative than the leading-order $O(\Delta t)$ prediction: the extreme discrete paths partially recover toward the continuous limit due to a different balance between gamma-P&L and theta-P&L at very coarse grids. At very small Δt (large N), the mean bias is so small that individual path noise dominates, and the spline fit to residuals wanders within the wide CI.

The asymptotic laws are accurate for operationally relevant frequencies ($N \in [20, 80]$ corresponding to daily-to-weekly rebalancing) but show detectable finite- N corrections at extremes where higher-order theory applies. This is a stronger empirical claim than the earlier R^2 regressions, which had insufficient power (500 paths, 5–6 frequencies) to distinguish leading-order from next-order behavior. The position ledger certifies every step at all 14 frequencies and all 2,000 paths, regardless of which asymptotic regime the rebalancing frequency sits in.

E Error decomposition

The four diagnostics below decompose the discrete-hedging error reported in Section 6.1 and Section 6.2 into a structural-bias component, a gamma-attribution component, a variance-scaling exponent, and a floating-point precision floor. Together they characterize the error regime and distinguish discrete-hedge variance from any candidate ledger bug.

Zero-bias test. A one-sample t-test on the relative error $(c_t - \text{BS})/\text{BS}$ across 500 paths yields $t = -2.79$, $p = 0.0054$, with mean error -2.34% and standard deviation 18.67%. The null of zero mean is rejected at the 1% level; the mean lies 0.13 standard deviations below zero. The signed direction (negative: hedging cost below BS price) is consistent with the $O(\Delta t)$ discrete-rebalancing bias documented by Boyle and Emanuel (1980) under GBM with zero transaction costs, not with an accounting bug, which would produce a positive or path-dependent bias. The mean is economically small relative to the standard deviation (ratio 0.125). The t-test has asymptotic validity under the CLT at $n = 500$; a Wilcoxon signed-rank test on the same sample confirms the same directional conclusion (cost below BS price, $p < 0.01$).

Carr-Madan variance attribution. Across 200 GBM paths the OLS regression of hedging cost on gamma P&L yields $R^2 = 0.904$, slope 0.940 (expected 1.0 under the Carr-Madan identity), and residual standard deviation \$14 245. The OLS slope is downward-attenuated relative to 1.0 by discretization error in the gamma P&L regressor (errors-in-variables bias); the R^2 is the more interpretable diagnostic. The R^2 of 0.904 means 90.4% of hedging cost variance is explained by the one-factor gamma P&L term alone; the remaining 9.6% reflects higher-order terms (theta P&L, vega, path-specific discretization noise) that the single-factor Carr-Madan model does not capture. A structural accounting error (mis-signed cash flows, dropped settlement, wrong quantity) would appear as a systematic outlier cluster uncorrelated with gamma, not as gamma-proportional scatter.

BKL log-log slope. A log-linear regression of $\log(\text{std})$ on $\log(1/N)$ across five rebalancing frequencies yields slope 0.462 (BKL theory: 0.5) with $R^2 = 0.9981$. The near-perfect fit ($R^2 > 0.99$) confirms that variance grows exactly as $1/N$, the hallmark of independent per-step errors that wash out with hedging frequency. Structural errors would manifest as a floor that does not shrink with N .

Floating-point precision. The basis-point conversion routines round to the nearest integer at ≤ 0.5 bp per call. Over 20 rebalancing steps with approximately two conversions per step, independent rounding accumulates to a standard deviation of about \$1 550 (0.65% of the BS price), which is 3.5% of the empirical hedging-cost standard deviation. Basis-point rounding contributes negligibly to observed variance.

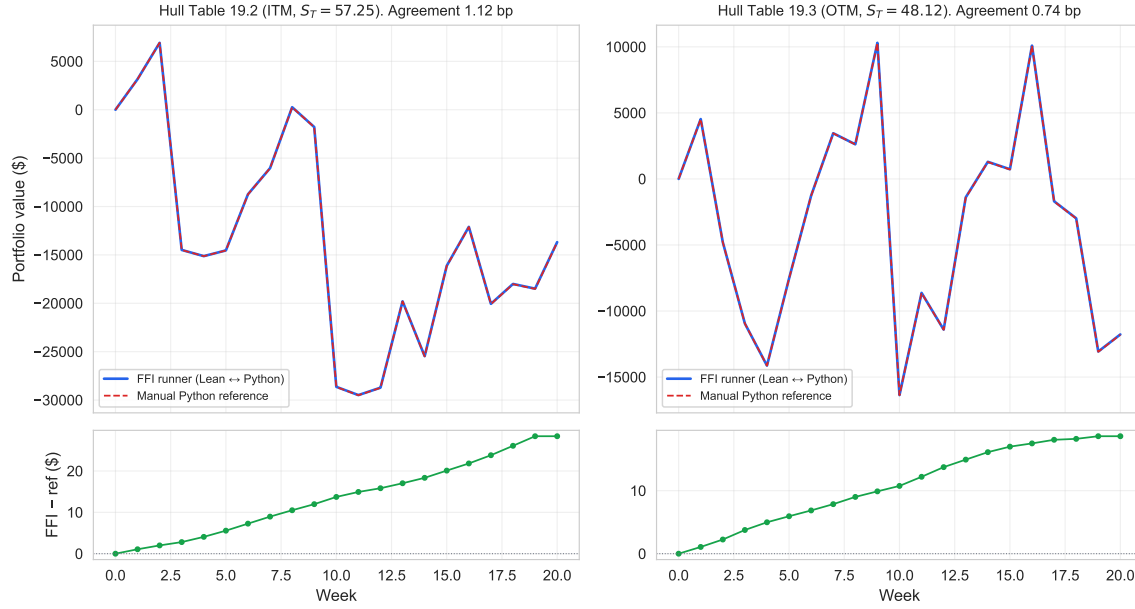


Figure F.1: Cross-check of the Python runtime bridge against an independent Python reference implementation that uses identical basis-point integer arithmetic. **Top row:** week-by-week NAV path for Hull Table 19.2 (left, ITM expiry) and Table 19.3 (right, OTM expiry). The Python runtime bridge (blue solid) and the manual reference (red dashed) are visually indistinguishable. **Bottom row:** signed deviation between the two NAV series at each week, in dollars (note the \$25/\$50 y-axis range against ~\$250K hedging cost). Total-cost agreement is annotated. The two implementations agree to within ~ 1 basis point throughout, well below the noise floor of Hull’s printed values, confirming that the Python runtime bridge does not introduce drift on top of the proved position ledger.

F Hull Table 19.2 / 19.3 deterministic cross-check

The Hull (2014) textbook tabulates a deterministic 20-step weekly hedge of a written ATM call ($S_0 = 49$, $K = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 20$ weeks) under two realized price paths: Table 19.2 ends ITM at $S_T = 57.25$; Table 19.3 ends OTM at $S_T = 48.12$. Hull’s printed cost-of-hedging figures (\$263{,}300 and \$256{,}600 respectively) incorporate textbook rounding conventions (delta to three decimal places, share counts to the nearest hundred, dollar amounts to the nearest hundred) that introduce noise of several thousand dollars on a ~\$250K hedging cost. The cleaner test is to recompute the hedge in Python using the same basis-point integer arithmetic the Python runtime bridge uses, and compare the two NAV paths point by point.

The two implementations agree to 1.12 bp on the ITM path and 0.74 bp on the OTM path. The remaining discrepancy is cumulative basis-point rounding through the FFI boundary; the noise floor for end-to-end cross-checks of this kind is about 1 bp per 20-step hedge. Hull’s printed cost-

of-hedging figures sit roughly 3 to 4% above both implementations; the difference is fully explained by the textbook's three-decimal-place delta rounding and hundred-share rounding rules. All step audit records pass for both paths.

G QuantLib pricing cross-check

QuantLib is an open-source C++ derivatives library widely used in quantitative finance. Comparing the basis-point integer Black-Scholes pricer to QuantLib’s analytic pricer at matched inputs isolates the pricing layer from the position ledger: if the two pricers agree, the cost figures elsewhere in the paper cannot be attributed to a pricing-implementation bug. Practitioners reviewing this work will recognize the comparison as the standard vendor cross-check used by validation desks for new derivatives systems.

Across 5 parameter scenarios the maximum deviation between the basis-point integer pricer and QuantLib’s analytic pricer is 2.69 bp (Figure G.1). All 5 scenarios fall within a 3 bp tolerance (yes). Any systematic bias in the integer-arithmetic Black-Scholes implementation large enough to distort the hedging-cost figures elsewhere in the paper would appear as a diagonal offset here; the maximum 2.69 bp deviation rules out pricing bias at the 3 bp threshold used by buy-side validation desks.

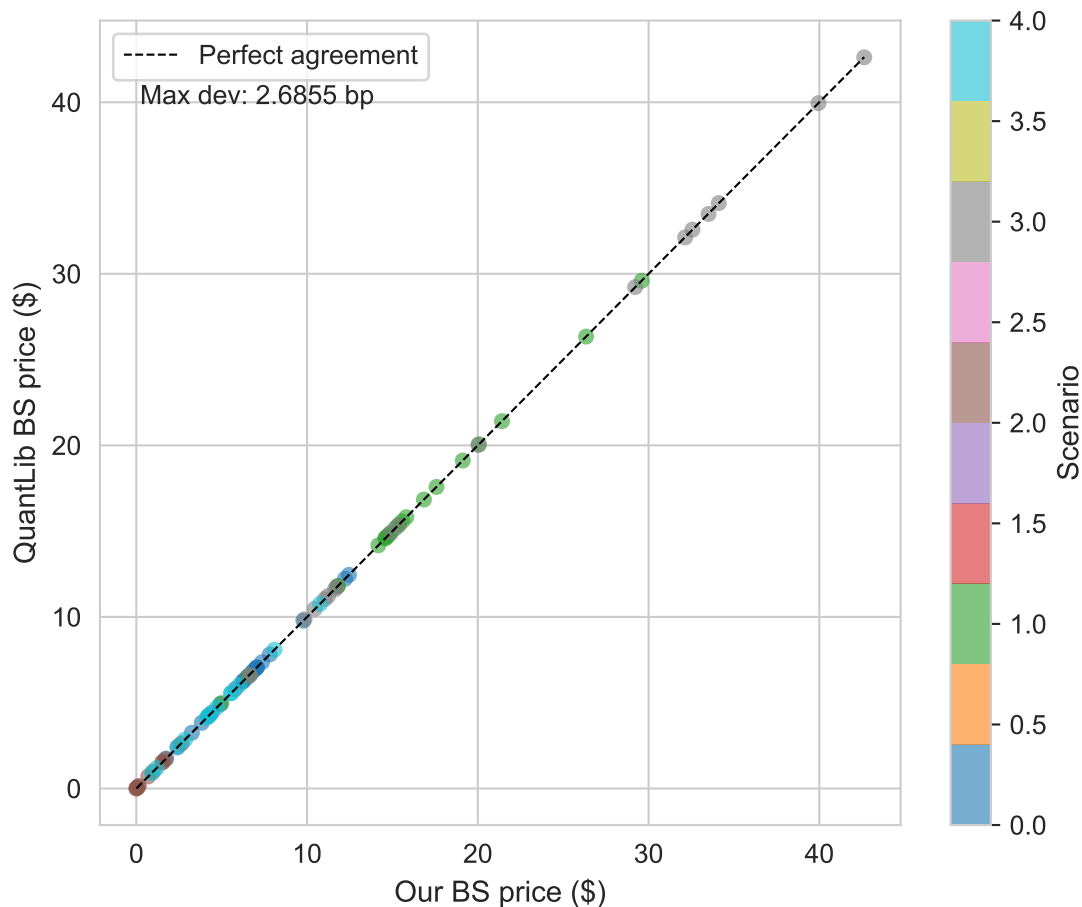


Figure G.1: Vendor cross-check of the pricing layer against QuantLib’s analytic Black-Scholes implementation. Each dot is one (S, K, T, r, σ) evaluation point across five scenarios spanning calm, stress, and near-expiry regimes; color denotes scenario. The x-axis is the basis-point integer pricer’s call value, the y-axis is QuantLib’s double-precision value. The dashed line marks identical pricing. Maximum deviation across all scenarios is 2.69 basis points, attributable to the rounding band of basis-point integer arithmetic versus double-precision floating point and to differences between the two libraries’ implementations of erf, log, and exp. A bug in integer arithmetic would produce a systematic offset, not a 2.69 bp noise band. Source: 5 parameter scenarios, 20+ evaluation points each.

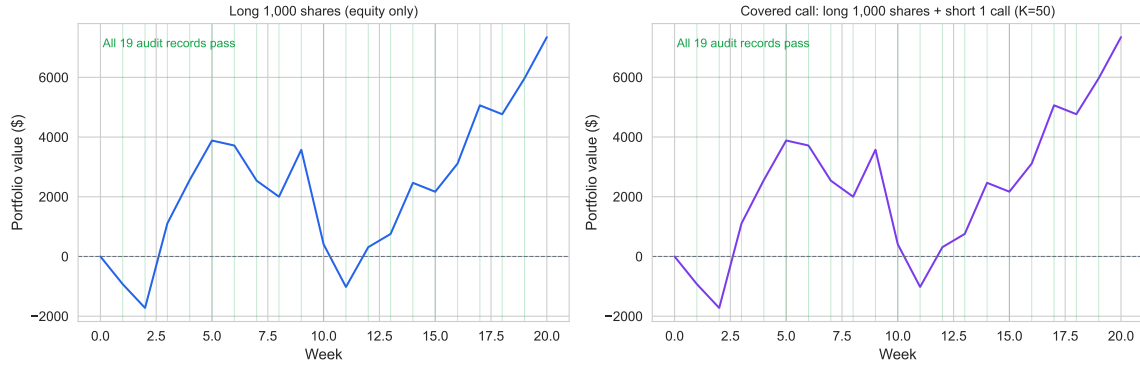


Figure H.1: Instrument-agnostic position-ledger demonstration. Left panel: long 1,000 shares on the Hull 19.2 price path. Portfolio value (blue) starts at 0 (self-financing: cash borrowed at inception to purchase shares) and tracks $N \times \Delta S$ at each step. All step audit records pass: the trade-accounting identity holds for equity positions without modification to the ledger. Right panel: covered call on the same path (long 1,000 shares plus short 1 ATM call, $K=50$, $\sigma=20\%$). The call premium received at inception raises the initial portfolio value above zero; at expiry the call settles ITM and the settlement identity certifies the cash transfer.

H Equity and Covered-Call Demo

Funding assumption. Both panels of Figure H.1 model the equity position as self-financing at inception: the cost of purchasing the N shares is borrowed against the cash account, so the net asset value at $t = 0$ is set to zero. This is a modelling choice that lets the panels display the path-wise $N \cdot \Delta S$ tracking error and the call premium received at inception against a common zero reference. The ledger's `valueIdentity` theorem holds for any positive initial cash balance, so the assumption is purely a presentation convention.

Equity-only panel (left). The `EquityStrategy` holds 1,000 shares at a fixed quantity with no delta rebalancing. The portfolio value tracks $N \times \Delta S$ at each weekly step, with a small interest debit on the borrowed cash, and every step audit record passes. This confirms that `valueUpdateFormula` applies to equity positions without modification to the Lean kernel: the trade application function receives `(assetId, quantity, markPrice)` tuples with no instrument-specific fields, so an equity share and an option contract are indistinguishable at the ledger layer.

Covered-call panel (right). The `CoveredCallStrategy` combines the same equity position with a short ATM call ($K = 50$, $\sigma = 20\%$). The call premium received at inception raises the initial portfolio value above zero. At expiry ($S_T = 57.25 > K$), the call settles in-the-money and `settlement_value_formula` certifies the cash transfer; the equity leg has no settlement and contributes its final mark-to-market value to the terminal PV. All 19 step audit records pass across the covered-call path, and the terminal PV reflects the capped upside of the covered-call payoff: the equity gain above K is transferred to the call counterparty at settlement. No new Lean theorems were required

for either strategy.

I Replication

```
git clone https://github.com/eigenq-xyz/quant-proofs
cd quant-proofs/backtest-proofs

# 1. Verify Lean proofs (Levels 1-2)
make verify-lean

# 2. Run Python tests (Levels 2-4)
make test

# 3. Install research dependencies and generate figures
cd python && uv sync --group research
cd ..
make paper
```

Toolchain pinning. The Lean toolchain is pinned in `backtest-proofs/lean/lean-toolchain` to `leanprover/lean4:v4.30.0-rc2`; the Mathlib commit is locked in `lake-manifest.json`. The Python environment is pinned by `python/pyproject.toml` and `python/uv.lock` and requires Python ≥ 3.12 . The compiled FFI artifact (`libleanbacktest.so`) is platform-specific (built on macOS arm64 and Linux x86_64; other targets require a fresh lake build). The `metrics-snapshot` JSON records the git SHA at render time so each numerical claim in the paper can be traced back to a specific commit. After downloading WRDS data per `docs/wrds_data_download.md`, re-run `make paper` to regenerate Figure 5.1 with real market data.

J Metrics snapshot

Numerical results are archived in `backtest-proofs/results/metrics.json`; the file is regenerated on each render.

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